



TERM 3

SYSTEM SOFTWARE AND COMPUTER MANAGEMENT

CHAPTER

6

CHAPTER OVERVIEW

Unit 6.1	Computer management
Unit 6.2	The role of the operating system
Unit 6.3	Management of files
Unit 6.4	Troubleshooting your computer
Unit 6.5	Factors that influence performance

Learning outcomes

At the end of this chapter you should be able to:

- Describe the role of the operating system.
- Discuss file management.
- Identify and explain the uses of operating system utilities.
- List the factors that influence computer performance.

INTRODUCTION

In this chapter, we will look at what an operating system is and why it is important in managing programs. We will also look at how to best manage files and which operating utilities you can use to improve performance. Finally, we will look at some of the factors that may decrease a computer's performance, as well as some potential solutions.

6.1 Computer management

In Grade 10, you learned about file management and how you can use files and folders in order to manage and order your documents.

FILES AND FOLDERS

Files refer to any document that you have created; including text documents, spreadsheets, images and music. These can be distinguished from one another by looking at the file extension. The **file extension** consists of the letters that follow after the full stop in the file name.

Let's take a look at some examples of file extensions:

- **.txt** for text files created in Notepad or other text editors
- **.docx** (or **doc**) for files created in Microsoft Word
- **.xlsx** (or **xls**) for files created in Microsoft Excel
- **.pptx** (or **ppt**) for files created in Microsoft PowerPoint
- **.pdf** for PDF files that are opened in a PDF reader (such as Adobe Acrobat)
- **.html** or **htm** for web pages
- **.png, jpg, tiff** for images
- **.mp3** for music files
- **.mp4** for video files

Folders refer to the location where the files are stored on your computer. You can create, name, rename and sort folders, as and when needed. They are usually stored on your hard drive and can be transferred to a USB drive, a CD, DVD, or the internet.

In Grade 11, you learned how to find a specific file and how to obtain additional information about the file. To find a file, you can use your computer's search function. To search for a file, you can do the following:

1. Open the **Start** menu.
2. Type in "File Explorer" and click the **File Explorer** option. You should now see a new window open.
3. Click the **Quick Search** box and type in the name of the file for which you are looking. (The **Quick Search** box is located at the top right-hand part of the window.)
4. Select the correct file from the search results and open by double-clicking the filename.

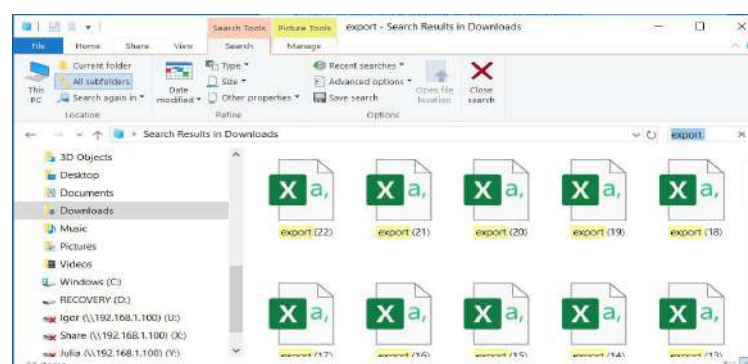


Figure 6.1: How to search for a file

FILE PROPERTIES

Once you have found the file you were looking for, you can obtain some additional information about the file by looking at its file properties. To open the *File Properties* window, you can do the following:

1. Open the folder where the file is saved.
2. Right click on the selected file to open the drop-down menu with options.
3. Scroll down the menu and select *Properties*.
4. A new window containing the file information will now open.

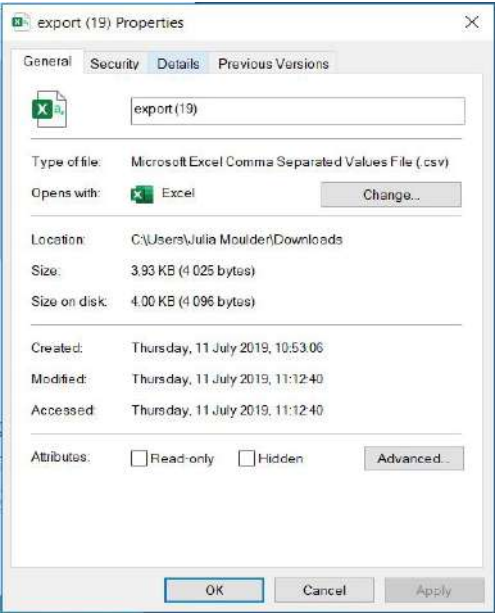


Figure 6.2: The File Properties window

You will notice that the *File Properties* window consists of the following four tabs:

1. *General*
2. *Security*
3. *Details*
4. *Previous versions*

Table 6.1: Tabs in the File Properties window

TAB	DESCRIPTION
<i>General</i>	The <i>General</i> tab contains information about the file type, the file location, the size of the document, and when the document was created, modified and accessed. It also indicates the attributes of the document, including <i>Read-only</i> and <i>Hidden</i> .
<i>Security</i>	The <i>Security</i> tab contains information on who has access to the document, as well as what they are allowed to do.
<i>Details</i>	The <i>Details</i> tab contains the file's meta information, which includes the title of the document and the name of the author.
<i>Previous versions</i>	The <i>Previous versions</i> tab indicates any previous versions of the document.



Example 6.1

The *Read-only* attribute allows the user to view, but not edit the file. In order to edit a read-only file, you need to change the file attribute and remove the read-only part. You can do this, by clearing the *Read-only* checkbox in the *File Properties* window.

The *Hidden* attribute makes the file invisible when you browse the folder. In order to view the file, you have to change the folder properties to allow you to view hidden files. This can be done as follows:

1. Open the *Start* menu.
2. Type in “control panel” and click *Control Panel*. A new window will open.
3. Click and open the *File Explorer* options. Your computer will now open a new window.
4. Select the *View* tab.
5. Under *Advanced Settings*, click *Show hidden files, folders and drives*.
6. Click *OK*.



- Write down the correct answer for each of the following questions.
 - Which one of the following file extensions are NOT used when you type a report?
 - .txt
 - .docx
 - .png
 - .pdf
 - Which of the following file extensions are NOT part of the Microsoft Office package?
 - .txt
 - .docx
 - .xlsx
 - .pttx
 - Which attribute allows the user to view, but not edit the file?
 - Hidden
 - Read-only
 - Security
 - Sharing
 - Which of the following tabs is not found on the *File Properties* window?
 - General*
 - Security*
 - Details*
 - Properties*
- Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 The location where the files are stored on your computer	A. <i>File Properties</i>
2.2 You can obtain some additional information about the file	B. .gif
2.3 Any document that you have created; including text documents, spreadsheets, images and music	C. .psd
2.4 File extension for an animation	D. Folder
2.5 File extension for a picture	E. .mov
	F. File
	G. .html
	H. .jpeg

3. Answer the following questions:
 - a. Define the concept of computer management.
 - b. List the positive outcomes of computer management.
 - c. Describe the four tabs visible when the *File Properties* window is displayed.
 - d. You want to hide a folder on the school's PC. List the steps that you will take in order to hide the folder.
 - e. How do you change the attribute of a file to enable yourself to edit the file?

6.2 The role of the operating system

An operating system is a group of system software that is responsible for controlling a computer's hardware and software, providing a user interface with which to interact and manage the computer's resources. Due to this, the operating system has a large number of different functions and jobs. To make these functions easier to understand, they will be grouped into different categories based on their goals.

These categories consist of the following:

- Starting the computer
- Providing a user interface
- Managing programs

STARTING THE COMPUTER

In computing, booting is the process of starting a computer. It can be initiated by hardware such as a button press, or by a software command. After it is switched on, a computer's central processing unit (CPU) has no software in its main memory, so some process must load software. It consists of picture-like items (icons and arrows for example). ...

GUI

The main pieces of a GUI are a pointer, icons, windows, menus, scroll bars, and an intuitive input device. Some common GUIs are the ones associated with Microsoft Windows, Mac OSX, Chrome OS, GNOME, KDE, and Android. It is loaded into memory before it can be executed.

PROVIDING A USER INTERFACE

In the past, you needed to know a lot about computers in order to open something as simple as a word-processing program. Fortunately, that has all changed. Today, even toddlers can use computers to open and play their favourite games. This is all thanks to the operating system.

The operating system makes the users' lives much easier by providing them with a user interface that can be used to interact with the computer. Most modern computers, laptops and smartphones use a graphical user interface (GUI), which allows the user to interact with programs using point-and-click technology.

Examples of a GUI include the following:

- Icons
- Program windows
- Menus
- Tabs on your browser
- Buttons



Figure 6.3: The GUI on a smartphone



Something to know

Microsoft's Windows 10 operating system is estimated to be made up of around 50 000 000 lines of code. In other words, it is made up of 50 000 000 lines of instructions that tell it how to do its job.



WHAT HAPPENS WHEN COMPUTER IS TURNED ON (BOOT PROCESS)



https://www.youtube.com/watch?v=Lp_Y_-DsfU

MANAGING PROGRAMS

One of the most important tasks of the operating system is to help the computer manage hundreds of processes that run at the same time. In other words, it needs to ensure that the appropriate number of resources is allocated to the processes, as well as the correct hardware is activated when required. An example of this can be seen when listening to music while you are playing games. You want your operating system to allocate the exact number of resources to each of the applications, allowing them to run at the same time. The last thing you want is for your computer to spend half the time playing music and the other half running your game. Fortunately, modern computers are very good at using resources effectively.

It should be noted that there are some circumstances where your computer might run out of resources. This includes running multiple high-intensity processes at the same time. When this happens, close other programs that are not important, or restart your computer. If the problems persist, you may have to buy a more powerful computer with more resources.



Figure 6.4: *Managing programs*

Something to know

A single-user operating system will allow you to create multiple user profiles. However, it will still be limited to only one active user.

SINGLE VERSUS MULTIPLE USERS

Operating systems can be designed with one or multiple users in mind. Single-user operating systems are normally designed for home and private use, as these computers use software that is licensed for a single user only. This means that only a single person can use the computer at one time. Examples of single-user operating systems include Microsoft Windows and MacOS.

Multiple-user operating systems allow more than one user to access and use the computer system at the same time. Because of this, these operating systems are normally used as server. The **server computer** can be accessed and used by multiple users through a computer network. A mainframe computer uses multiple computer terminals that are connected to the main computer. Examples of multi-user operating systems include Unix and Linux.

MULTITASKING

One drawback of older operating systems is that they were only able to load one program onto the computer's memory at a time. This means that the computer could only run a single program at a time.

Multitasking operating systems allow the computer to run multiple processes (tasks), or applications at the same time. An example of this can be seen when you use your smartphone. You can listen to music and at the same time, browse the internet, or chat to a friend on WhatsApp. This is *not* where multitasking stops, however; it also allows certain services to run in the background. These background services allow your smartphone to check for messages, update your applications and manage your cellular network all at the same time.

TASK MANAGER (WINDOWS)

In order to find out how a computer is using its resources, Windows operating systems have been designed to include an application called *Task Manager*. *Task Manager* allows the user to see which tasks and applications are currently running on the computer, and how much memory and CPU processing power is being dedicated to each individual task. It also allows the user to forcefully end any applications that might have stopped working, or are using too many resources. The *Task Manager* window consists of several tabs; each dedicated to provide the user with specific information. Table 6.2 lists the description for each of these tabs.

Table 6.2: *Tabs in the Task Manager window*

TAB	DESCRIPTION
<i>Processes</i>	The <i>Processes</i> tab gives the user information on which programs are running and the resources that they are using.
<i>Performance</i>	The <i>Performance</i> tab contains information on how your hardware is being used. This includes your CPU, memory, hard drive, network and GPU.
<i>App History</i>	The <i>App History</i> tab shows the user how much of your CPU and network have been used by the various applications. This is measured from the date indicated to current.
<i>Start-up</i>	The <i>Start-up</i> tab provides information on which applications automatically become active once your computer is turned on. Disabling some applications can speed up the boot process.
<i>Users</i>	The <i>Users</i> tab provides more information on which users are currently signed into the computer and how many resources they are using.
<i>Details</i>	The <i>Details</i> tab displays every process that is currently running on the computer.
<i>Services</i>	The <i>Services</i> tab provides information on all the Windows services that have been installed on your computer.

To open the *Task Manager*, you can do the following:

1. Open the *Start* menu and enter the words "task manager".
2. Click on *Task Manager*.



Something to know

When using a multiple-user operating system, it is important to make sure that you balance the requirements of each user so that the resources of the main computer are shared among the users.



Something to know

One of the reasons why your smartphone's battery goes flat, even when the screen is turned off and you are not using it, is that your CPU is still running tasks. This includes checking for updates, looking for and downloading messages and emails, and making sure you are connected to an internet-enabled network.

On most Windows 10 computers, there will be more than 50 applications and processes running at any given time.

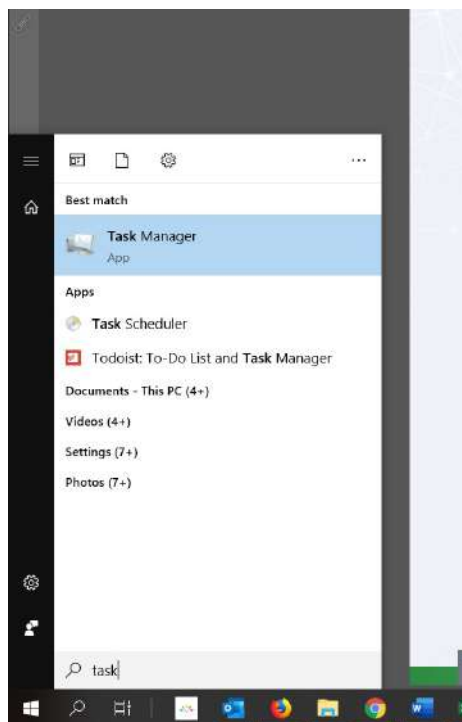


Figure 6.5: Opening Task Manager in the Start menu



Activity 6.2

1. Write down the correct answer for each of the following questions.
 - a. Which tab provides information on the applications that automatically become active once your computer is turned on?
 - A. Start-up tab
 - B. User tab
 - C. Process tab
 - D. Performance tab
 - b. Which tab provides more information on which users are currently signed in to the computer and how many resources they are using?
 - A. Performance tab
 - B. Start-up tab
 - C. User tab
 - D. Process tab
 - c. Which of the following is NOT a GUI?
 - A. Menus
 - B. Buttons
 - C. Icons
 - D. Backlight
 - d. Which of the following is a shortcut used to open Task Manager?
 - A. Alt+F4
 - B. Ctrl+Shift+Esc
 - C. Ctrl+Shift+T
 - D. Ctrl+Shift+D



Activity 6.2

- e. Which application do you use to find out how the computer is using its resources?
- A. Task Manager
 - B. Settings
 - C. Windows Performance
 - D. Device Manager
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Contains information on how your hardware is being used, which includes your CPU, memory, hard drive, network and GPU	A. <i>Details</i> tab
2.2 Gives the user information on which programs are running and the resources that they are using	B. <i>Services</i> tab
2.3 Displays every process that is currently running on the computer	C. <i>Performance</i> tab
2.4 Shows the user how much of the CPU and network have been used by the various applications	D. <i>Coke and Tab</i>
2.5 Provides information on all the Windows services that have been installed on your computer	E. <i>Process</i> tab
	F. <i>Thinking</i> tab
	G. <i>App History</i> tab
	H. <i>Users</i> tab

3. Answer the following questions:
- a. Explain the role of the operating system.
 - b. List the three categories into which the functions of the operating system are divided.
 - c. Provide two ways in which to open *Task Manager*.
 - d. Define “multitasking” in terms of a computer operating system.
 - e. Discuss the difference of single user versus multiple users. Give a user type and a disadvantage that is associated with each.



File management ensures that a computer is user friendly, that no files go missing, and that you will be able to find your data and information without wasting any time.

FILE MANAGEMENT

File management allows users to name, organise and arrange their files and folders in such a way that they are easy to find and distinguish. In order to help you with this process, let's take a look at some tips on effective file management:

- When installing new programs, use the default installation folders. This will ensure that all your programs are installed in the same folder.
- Save all your documents in the same root folder. This also applies to all music (music folder), pictures (images folder) and videos (videos folder).
- Create subfolders within the root folder in order to distinguish between documents.
- Name or rename your folders appropriately. When a folder is created, it will be named "New Folder". Rename the folder according to the contents of it.
- File new documents as they are created. This will ensure that the document is in the correct folder and, therefore, can be found with more ease.
- Remove old documents and folders. Old documents and folders can take up space and cause your computer to become cluttered. In order to prevent this, you should delete documents and folders that are no longer needed. If you do not want to delete them, consider creating a root folder called "oldies" and moving all old documents to that folder.
- Make regular backups of important files. This will prevent you from losing any important information.

IMPORTING AND EXPORTING

Modern operating systems allow you to import and export files, documents and data. Importing a file refers to bringing a file from a different program into the one that you are currently using. An example of this is when you import information from a Microsoft Excel (.xls) file to Microsoft Word (.doc). Exporting a file refers to saving the file in a way so that it can be used by a different program. This normally includes converting the file format to one that is usable by the program. An example of this is exporting a file from Microsoft Word (.doc) to a PDF reader (.pdf).

Importing and exporting can also be used to manage files and documents on your computer or smartphone. This is best illustrated when exporting your contacts from your smartphone and transferring the data to your Google account (this can also be used to back-up the data). Once the data has been uploaded, you can import the contacts to your computer, tablet, or new smartphone.



Activity 6.3

1. Briefly explain the following
 - a. Spooling
 - b. Firewall
 - c. Spoofing
 - d. File compressors
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Allows users to name, organise and arrange their files and folders in such a way that they are easy to find and distinguish	A. Antivirus
2.2 A utility that enables a printer to print documents that are much larger than the amount of memory that the printer even has	B. File management utilities
2.3 Programs that use advanced algorithms and mathematical formulas to decrease the amount of space used by files	C. Importing
2.4 Utility that provides users with a GUI that they can use to organise files and folders	D. File management
2.5 Refers to bringing a file from a different program into the one you are currently using	E. Firewall
	F. Spooling
	G. Exporting
	H. File compression

3. Answer the following questions:
 - a. Give two main reasons why it is good to constantly keep your computer programs updated.
 - b. List two reasons why file compression is rarely used.
 - c. Explain “converting” and give an example of when it is implemented.



OPERATING SYSTEM UTILITIES

Most operating systems include various system utilities. These utilities are small programs that work with the operating system in order to help users maintain and protect their computers. The following are some of the most commonly used utilities:

- Firewall
- File compressors
- File management utilities
- Update schedulers
- Backups

FILE COMPRESSORS

File-compression programs use advanced algorithms and mathematical formulas to decrease the amount of space used by files. While this sounds like a great way to free storage space, in reality, file compression is rarely used to free storage space. There are two important reasons for this:

1. Compressed files **cannot** be used immediately; they first need to be decompressed. Not only does this waste time every time you open up a file; it uses up the space you saved when you compressed the files in the first place.
2. The files most likely to use up your storage space are already compressed. This includes most video and music files. As such, compressing them further does not help.

Compression is, however, useful to archive files. When many files and folders are grouped together into a single file, they are said to be archived. This makes them easier to store or transfer. For example, it is a lot easier to email a single archived file than thousands of separate files. Since a single compressed file can contain thousands of files and folders, they are often used to archive files.

FILE MANAGEMENT UTILITIES

File management utilities provide users with a GUI that they can use to organise files and folders. Operating systems, such as Windows, use a variety of different GUI options to enable users to customise files and folders to best suit their needs.

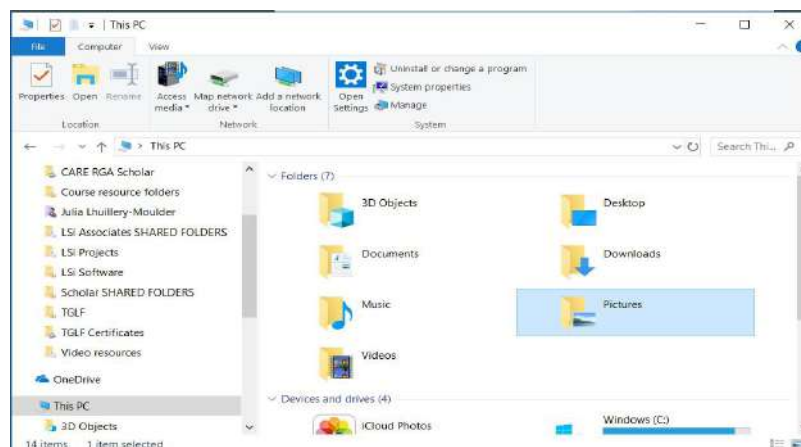


Figure 6.6: *My Computer is an example of a file management utility*

SCHEDULING UTILITY

In today's competitive market, computer programmers continue to improve their software after they have been released. This includes adding new features, fixing bugs, improving system reliability or security. In order to obtain these improvements, the user needs to update the software, or download a patch from the supplier's website.

If an application or operating system asks you if it can automatically install an update, the best idea is to simply allow it to do so. This takes a few minutes to complete, but ensures that your computer is up to date with the newest software. Since updates are so important for software, saying no to an update will cause the application to constantly remind (and annoy) you about the update, until you eventually allow it. As such, it is better to update the application as soon as possible.

The Scheduling Utility is located in the System ribbon menu. The new Scheduling feature provides savings in both time and human resources and greatly improves the efficiency with which you can automate repetitive and time consuming activities. The three areas controlled by this utility are Report Scheduling, Email Alerts and Process Scheduling.

BACKUP UTILITIES

Operating systems, such as Windows, come equipped with a backup and restore utility. This program allows users to back up important files and folders, as well the important settings for the operating system. To use *Backup and Restore*, you can do the following:

1. Purchase an external hard drive to use for the backup and connect it to your computer.
2. Open the *Start* menu and type in "control panel". This will open the *Control Panel*.
3. Click *Backup and Restore* (in Windows 7) to open the *Backup and Restore* window.
4. Click *Set up backup*.
5. Select the external hard drive as the location where you would like to save the backup.
6. If all your files are stored in your private folders (such as the Documents, Music and Videos folders), let Windows choose. Otherwise, select the folders that should be backed up manually.
7. Click *Save and run the backup* option to create a backup of all your files.

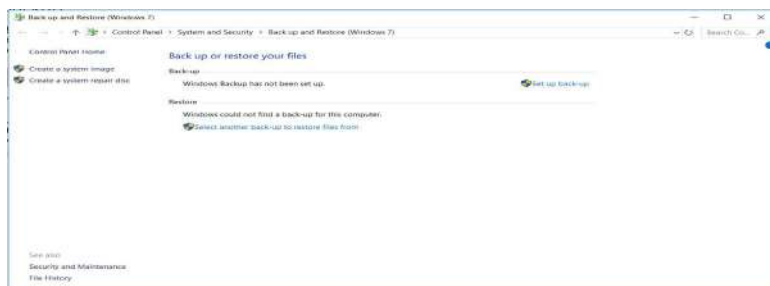


Figure 6.7: Creating a backup using the Backup and Restore application



To restore files from a Windows backup, you can do the following:

1. Connect your external hard drive to your computer.
2. Open the *Backup and Restore* (Windows 7) application from the *Control Panel*.
3. Click *Restore all users' files*.
4. Click *Browse for files* or *Browse for folders*.
5. Select in the original location box and click *Restore*.
6. This will restore your files to where they used to be; therefore, the files are no longer deleted.

DEFRAGMENTATION

When you save a file onto your computer, the computer breaks the file up into smaller pieces. This is done so that the file can be stored on your hard drive. Unfortunately, the file pieces are not always stored in the same location, but rather spread over several locations. When this happens, it is known as fragmentation.

Fragmentation occurs from files being constantly written, deleted and resized, and can lead to several problems. These include a slow computer, long boot-up times, random computer crashes and the computer freezing. In order to prevent these problems from happening, you can defragment your hard drive. This process reorganises your hard drive by taking all the pieces of data that are spread across your hard drive and reorganising it so that the pieces are closer to each other.

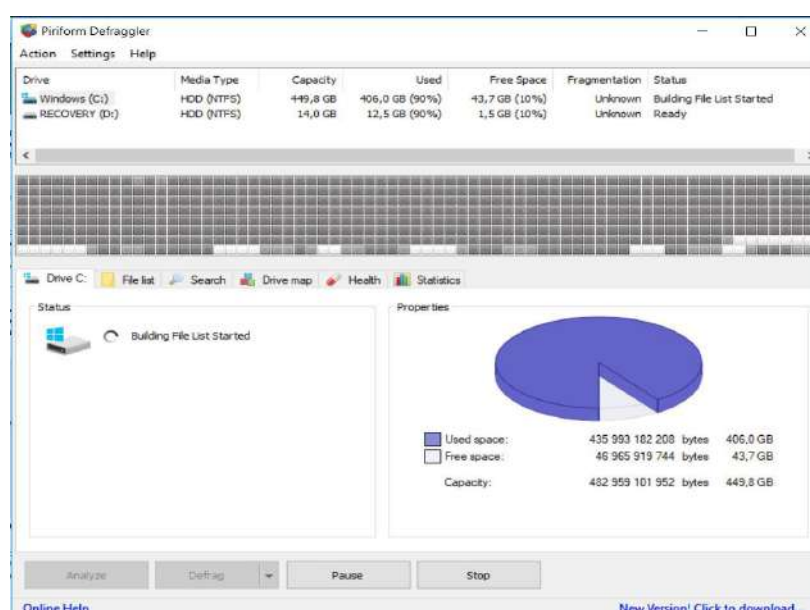


Figure 6.8: The process of fragmentation and defragmentation

Examples of programs that you can use to defragment your computer include the following:

- *Optimise Drive Tool*, which comes with the Windows operating system and was previously known as *Disk Defragmenter*
- PerfectDisk
- Norton Utilities



SPOOLING

Spooling is a process in which data is temporarily held to be used and executed by a device, program or the system. Data is sent to and stored in memory or other volatile storage until the program or computer requests it for execution.

“Spool” is technically an acronym for simultaneous peripheral operations online.

Spooling works like a typical request queue or spool where data, instructions and processes from multiple sources are accumulated for execution later on. Generally, the spool is maintained on the computer’s physical memory, buffers or the I/O device-specific interrupts. The spool is processed in ascending order, working on the basis of a FIFO (first in, first out) algorithm.

The most common implementation of spooling can be found in typical input/output devices such as the keyboard, mouse and printer. For example, in printer spooling, the documents/ files that are sent to the printer are first stored in the memory or printer spooler. Once the printer is ready, it fetches the data from that spool and prints it.



Activity 6.4

1. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
1.1 System utilities that come standard with an operating system that helps to find computer-related problems	A. Fragmentation
1.2 Occurs due to files being constantly written, deleted and resized, and can lead to several problems	B. Disk Clean-up
1.3 The tool that identifies different types of files that can be deleted without affecting the user	C. Troubleshooting

2. Answer the following questions:
 - a. Explain the process of fragmentation and defragmentation.
 - b. Explain how a troubleshooting wizard works.

6.4 Troubleshooting your computer

RUN DISK CLEANUP

To show the *Disk Clean-up* process, watch the following video:



<https://www.youtube.com/watch?v=5UfMKd3SPtY>

In Chapter 1, we looked at some of the potential problems (and solutions) that you might encounter while using a computer. So, in order to make your life even easier, in this section, we will be taking a look at some of the utility software that you can use in order to resolve your computer problems.

UTILITY SOFTWARE FOR TROUBLESHOOTING

DISK CLEAN-UP

Sooner or later, you are going to encounter the problem of running out of hard-drive storage space. When this occurs, the *Disk Clean-up* utility can be a very handy tool to free up some hard-drive space. The tool comes standard with Microsoft Windows and works by identifying different types of files that can be deleted without affecting you. These include temporary files, the files in your *Recycle Bin* and the files used as part of a completed software upgrade. Once the *Disk Clean-up* tool has identified the files, it lets you decide which files can be deleted.

To use the *Disk Clean-up* tool, you can do the following:

1. Click the *Start* menu.
2. Type in “disk clean-up” and select *Disk Clean-up*. Your computer will calculate how much space you can free up.

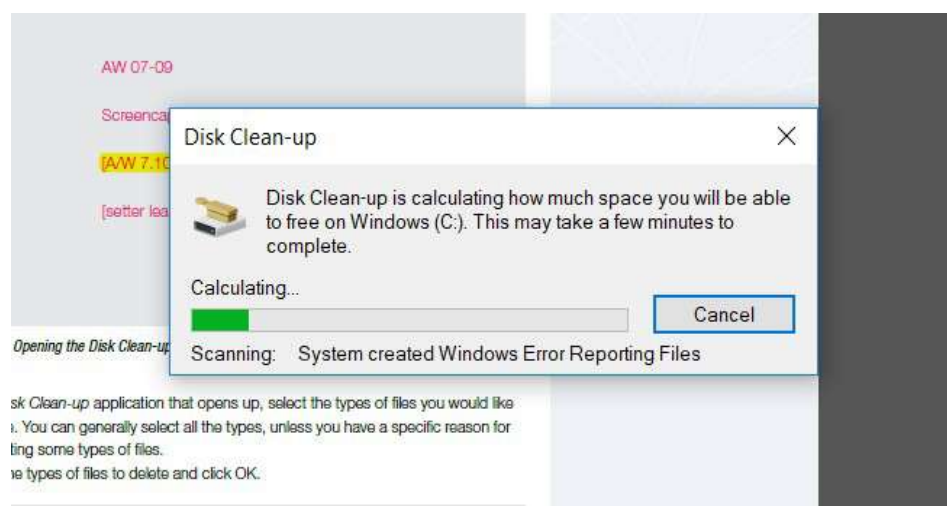


Figure 6.9: Opening the *Disk Clean-up* tool

3. In the *Disk Clean-up* application that opens up, select the types of files you would like to delete. You can generally select all the types, unless you have a specific reason for not deleting some types of files.
4. Select the types of files to delete and click OK.

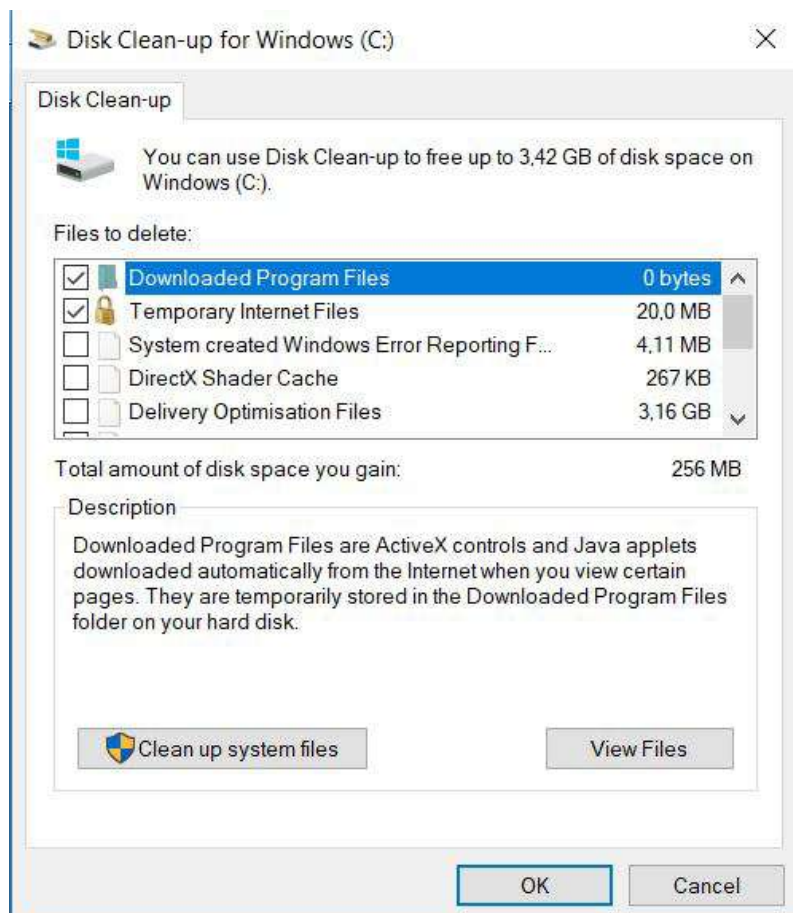


Figure 6.10: Using the Disk Clean-up tool

The amount of space cleaned up with this application will depend on how regularly you use it. If you rarely use the *Disk Clean-up* tool, it can free up a few gigabytes at a time. However, if you use it regularly, it may only free a few megabytes at a time.

WIZARDS

Troubleshooting can be a time-consuming and annoying procedure; fortunately, your operating system can fix this. Troubleshooting wizards are system utilities that come standard with your operating system. The tool allows you to quickly check if your operating system or hardware is functioning as required. If a problem is encountered, the wizard will attempt to fix it. If the wizard cannot fix the problem, it will ask you some questions in order to determine the cause of the problem.

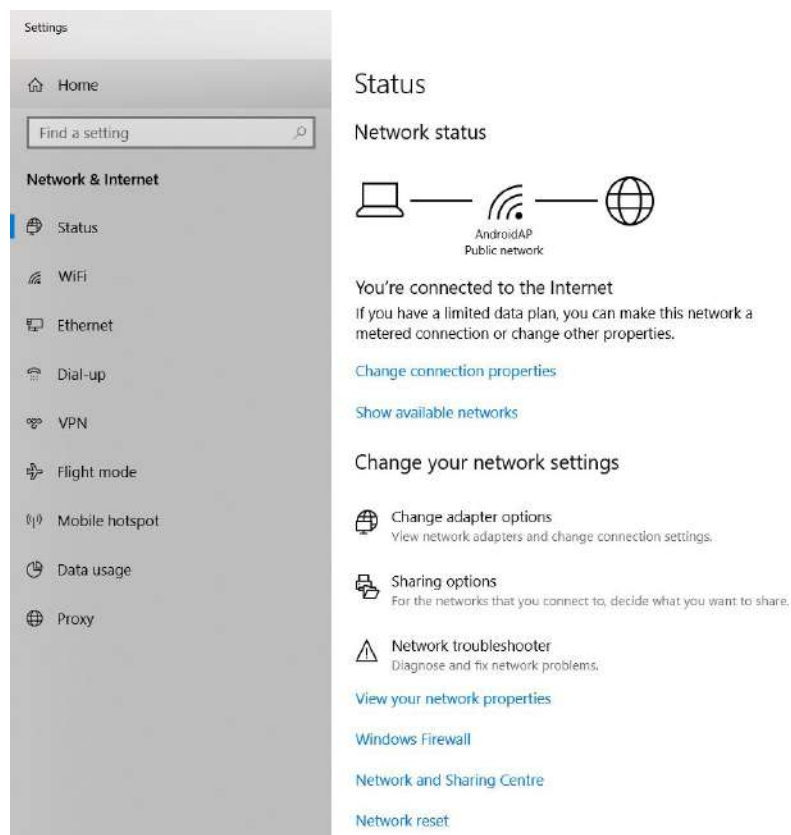


Figure 6.11: Using a wizard to fix a network connection

Examples of problems that can be fixed using the troubleshooting wizards include network connection problems, printing problems and software compatibility problems.

6.5 Factors that influence performance

As you are most likely aware, you use your computer a lot. Due to this, the computer is subject to some wear and tear, which can reduce its speed and performance. In order to help you determine *why* your computer is becoming slower, we will be taking a look at some of the most common factors that might influence computer performance.

RAM

RAM is responsible for temporarily storing the data that is being used by the computer's processor. This allows the processor to access the data much faster and easier than would be possible if the data was loaded from a secondary storage device, such as a hard drive.

A slow computer can be an indication that you have too little RAM. This can be solved by buying and adding additional RAM to your computer. If you are looking to buy additional RAM, do proper research regarding the speed and type of RAM that is used by your motherboard.

PROCESSOR

The CPU is the part of a computer that is responsible for receiving and carrying out a computer's instructions. Because of this, the CPU is one of the most important components for determining how fast a computer will be.

There are two main factors that influence the speed of the CPU:

1. The first is the GHz of the processor, which refers to the CPU frequency and is an indication of the processor speed. As a general guideline, the higher the frequency, the better the CPU.
2. The second factor is the amount of cores. As with processor speed, more cores are generally better.

APPLICATIONS

Every computer has a limited number of resources (such as RAM and CPU processing time) that can be used to run applications. For each application currently running, the amount of available resources will become less. If you open too many applications at the same time, all available resources will be used up and your computer's processing speed will slow down significantly. This might even lead to some applications freezing entirely.

In order to prevent this from happening, close any application that you are not currently using. You can do this by using the *Task Manager* application, as described earlier in this chapter.

CACHING

One of the methods that your computer uses in order to improve processing speed is caching. **Caching** is a small, temporary form of storage that is created in a fast form of



FIXING YOUR SLOW COMPUTER

To show why a computer becomes slow, watch the following video:



https://www.youtube.com/watch?v=k_RpftRwrNk



Something to know

More RAM allows more data to be stored in the RAM, which allows your programs to run faster.



memory in order to allow data to be accessed more quickly. An example of this can be seen when you are browsing the internet. When you open a website, such as Facebook, your computer temporarily stores the files for the website in your cache. Next time you open Facebook, the files are loaded from the cache instead of the internet. This not only helps the site to load faster; it also reduces the burden on your network.

Your computer uses the following three forms of caching:

1. **CPU cache:** The difference in speed between a computer's CPU and RAM has recently become too large. As a result, the RAM cannot provide all the information the CPU needs fast enough. To prevent this problem, many CPUs now have a small, high-speed cache built into the CPU, where it can temporarily store the data it needs.
2. **Disk cache:** The disk cache is a small amount of RAM that is built into a hard-disk drive. It stores data that is being sent or received by the hard-disk plates for a short period of time.
3. **Web cache:** The web cache is a small area on a computer's hard drive where images and pages from the world-wide web are stored for a short period. This allows you to quickly load those images from your computer when you visit the page again, rather than to wait to have them load over a slow internet connection.

DISK OPTIMISATION

As discussed previously in this chapter, fragmentation of data can cause your computer to slow down. To prevent this from happening, you can use a defragmentation tool to reorganise the data on your hard drive. This will rearrange the areas where data is stored on the hard drive to make sure the pieces of a file are close to each other and that the fastest parts of the hard drive are used.

Another potential problem that you might encounter is running out of storage space. When this happens, you have the following options:

- Delete any files that you do not need any more.
- Use the **Disk Clean-up** application to help create some additional space.
- Buy an additional hard drive.

MALWARE

As discussed in Chapter 5, malware has been designed to cause harm and use your computer's resources without your knowledge. Due to this, malware is often responsible for slowing down your computer. In order to prevent malware from infecting your computer, or to remove any malware that might already be installed on your computer, you can use security applications, such as a firewall, antivirus and anti-spyware.

A processor, or "microprocessor," is a small chip that resides in computers and other electronic devices. Its basic job is to receive input and provide the appropriate output. While this may seem like a simple task, modern processors can handle trillions of calculations per second.

Something to know

Computers are not limited to a single hard drive. If you are running low on storage space, do not hesitate to buy an additional hard drive to store more data.

Something to know

Windows 10 comes equipped with a basic firewall and antivirus program. If you feel that you will need additional protection, Chapter 6 has some recommendations on which programs to buy.



The central processor of a computer is also known as the CPU, or “central processing unit.” This processor handles all the basic system instructions, such as processing mouse and keyboard input and running applications. Most desktop computers contain a CPU developed by either Intel or AMD, both of which use the x86 processor architecture. Mobile devices, such as laptops and tablets may use Intel and AMD CPUs, but can also use specific mobile processors developed by companies like ARM or Apple.

Modern CPUs often include multiple processing cores, which work together to process instructions. While these “cores” are contained in one physical unit, they are actually individual processors. In fact, if you view your computer’s performance with a system monitoring utility like Windows Task Manager (Windows) or Activity Monitor (Mac OS X), you will see separate graphs for each processor.

Processors that include two cores are called dual-core processors, while those with four cores are called quad-core processors. Some high-end workstations contain multiple CPUs with multiple cores, allowing a single machine to have eight, twelve, or even more processing cores.

Besides the central processing unit, most desktop and laptop computers also include a GPU. This processor is specifically designed for rendering graphics that are output on a monitor. Desktop computers often have a video card that contains the GPU, while mobile devices usually contain a graphics chip that is integrated into the motherboard. By using separate processors for system and graphics processing, computers are able to handle graphic-intensive applications more efficiently.



Activity 6.5

1. Which of the following refers to cache located on the CPU?
 - a. Disk cache
 - b. CPU cache
 - c. Web cache
 - d. Motherboard cache
2. Answer the following questions:
 - a. Define RAM.
 - b. Explain what influences a CPU’s speed.
 - c. What can potentially happen if you open too many applications at the same time?
 - d. Describe how cache improves performance.
 - e. List three ways in which you can free up disk space.
 - f. What is meant by a CPU’s generation and how does this affect its speed?





REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

- 1.1 Where can you search for additional information about a file? (1)
A. *Properties*
B. Windows Explorer
C. *Security* tab
D. *Start* menu
- 1.2 Which of the following tabs is only found in the *Properties* window of a folder? (1)
A. General
B. Security
C. Details
D. Location
- 1.3 When a computer is started up the operating system is loaded into the _____. (1)
A. ROM
B. RAM
C. Hard disk
D. Cache
- 1.4 Which of the following is not on a smartphone's GUI? (1)
A. Browser tabs
B. Buttons
C. Settings menu
D. Mouse icon
- 1.5 Which of the following troubleshooting utilities is used to increase your computer's storage capacity? (1)
A. Defragmentation tool
B. Disk Clean-up tool
C. Wizard tool
D. Antivirus software

[5]

QUESTION 2: TRUE OR FALSE

Indicate if the following statements are TRUE or FALSE. Correct the statement if it is false. Change the underlined word(s) to make the statement true.

- 2.1 Multitasking operating systems allow more than one user to access and use the computer system at the same time. (1)
- 2.2 A new computer's performance increases over time. (1)
- 2.3 The write-only attribute allows the user to view but not edit the file. (1)
- 2.4 Print management allows a printer to print documents that are much smaller than the amount of memory the printer has. (1)
- 2.5 CPU cache is high-speed memory stored on CPU that is larger than RAM. (1)

[5]





QUESTION 3: MATCHING ITEMS

Choose a term or concept from Column B that matches a description in Column A.

COLUMN A	COLUMN B
3.1 A system that allows the computer to run multiple tasks or applications at the same time.	A. Performance tab B. Disk Clean-up tool C. Multiple user operating system
3.2 Where you go to see which tasks and applications are impacting the performance of your computer, so you can stop any unnecessary processes.	D. Single user operating system E. Previous version tab
3.3 Where you go to see which applications are automatically activated when your computer is turned on.	F. Task Manager G. App History tab
3.4 Where you go to find information on how your hardware is being used.	H. Disk cache I. General tab J. Wizard tool
3.5 A system that is used for a person's personal use. The software on this computer can only be used by one person at a time.	K. Multitasking operating system L. Startup tab M. Web cache N. Processes tab

[5]

QUESTION 4: MEDIUM AND LONG QUESTIONS

- 4.1** How can a wizard be used to improve the troubleshooting process? (3)
4.2 What is a multitasking operating system? (2)
4.3 Mention TWO situations where your computer can run out of resources. (2)
4.4 Mention ONE profession that would use a single user operating system. Give a reason for your answer. (2)
4.5 Mention ONE profession that would use a multiple user operating system. Give a reason for your answer. (2)

TOTAL: 21

AT THE END OF THE CHAPTER

NO.	CAN YOU ...	YES	NO
1.	Describe the role of the operating system?		
2.	Explain file management?		
3.	Identify and explain the uses of operating system utilities?		
4.	List the factors that influence computer performance?		



INTERNET COMMUNICATION
TOOLS

CHAPTER OVERVIEW

Unit 7.1	Electronic communication
Unit 7.2	Web browsers
Unit 7.3	Computer communications
Unit 7.4	Communication devices

By the end of this chapter, you will be able to:

- Explain what digital communication is.
- Identify and discuss the different types of digital communication.
- Describe the most common features of a web browser.
- Discuss some of the most popular communication devices.
- List and describe the uses of computer communications.

INTRODUCTION

In this chapter, we will take a look at how the internet and the world-wide web are used to connect people around the world, as well as the tools that make this possible. We will start by discussing digital communication and communication devices. Thereafter, we will identify and discuss some of the most common and useful features of web browsers. Finally, we will take a look at the some of the social implications related to digital communications.



Figure 7.1: *The world-wide web*

In previous years, you learned that digital or electronic communication refers to any data, information, words or photos that are sent electronically in order to communicate with one or more people. This includes calls, messages, group chats, emails, social networks and websites. These methods of electronic communication have made our lives much easier and enabled us to communicate with people around the world. However, they do have some flaws and risks that you need to be aware of.

ADVANTAGES OF ELECTRONIC COMMUNICATION

- **Speedy transmission:** It requires only a few seconds to communicate through electronic media because it supports quick transmission.
- **Wide coverage:** The world has become a global village and communication around the globe requires only a second.
- **Low cost:** Electronic communication saves time and money. For example, text SMS is cheaper than the traditional letter.
- **Exchange of feedback:** Electronic communication allows the instant exchange of feedback. So communication becomes perfect using electronic media.
- **Managing global operation:** Due to the advancement of electronic media, business managers can easily control operation across the globe. Video or teleconferencing e-mail and mobile communication are helping managers in this regard.

DISADVANTAGES OF ELECTRONIC COMMUNICATION

- **The volume of data:** The volume of telecommunication information is increasing at such a fast rate that business people are unable to absorb it within the relevant time limit.
- **The cost of development:** Electronic communication requires huge investment for infrastructural development. Frequent change in technology also demands further investment.
- **Legal status:** Data or information, if faxed, may be distorted and will cause zero value in the eye of law.
- **Undelivered data:** Data may not be retrieved due to system error or fault with the technology. Hence required service will be delayed.
- **Dependency:** Technology is changing every day and therefore poor countries face the problem as they cannot afford the new or advanced technology. Therefore poor countries need to be dependent towards developed countries for sharing global network.

ELECTRONIC COMMUNICATION BEST PRACTICES

When you communicate on the internet – whether it is via email, instant messaging, or posting a blog – it is important that you follow proper **netiquette**. This will not only make the internet a more pleasant place for everyone else, it will also save you from potential embarrassment in the future!



GUIDELINES WHEN COMMUNICATING ON THE INTERNET

- Texting (messaging):
 - Keep texts short
 - Longer texts can be misinterpreted
 - Sign a text with your name
 - Spell out all words and do not use “texting lingo” or shorthand
 - Texts can be saved and can be altered
- Email:
 - Use a descriptive subject line
 - Be courteous
 - Reply promptly – but allow yourself time to get over an initial reaction to an angry email
 - Remember attachments to an email may contain metadata that can disclose unwanted information to the recipient
- Social Media:
 - There is no expectation of privacy on the internet
 - Change your passwords frequently
 - Log off after visiting the page
 - Delete your browsing history, saved passwords and cookies regularly
 - Do not disparage anyone via social media
 - Educate yourself about a site before joining.

While the tips covered in this section are generally good guidelines, it is important to note that netiquette differs from site to site and changes over time. Find out what is acceptable behaviour on that website before sending your own messages.

TYPES OF ELECTRONIC COMMUNICATION

With the advances in computer technology and the internet, there are many new and exciting ways to communicate; from sending instant messages on social network sites, to email. In this section, we will discuss the most popular types of digital communications, their advantages, as well as their disadvantages.



Something to know

Asynchronous communication means that a message can be sent without the message receiver being available at the time. Most messaging applications are asynchronous. Most phone call applications, however, are synchronous.

EMAIL

Email is one of the first and most popular forms of electronic communication. It allows the user to send and receive files and messages over the internet, and can be used on a wide variety of devices. Here are some of the advantages and disadvantages of email.

ADVANTAGES OF EMAIL

- Email is a free tool.
- Email is quick. Once you have finished composing a message, sending it is as simple as clicking a button. Once it is sent and delivered, it can be read almost immediately.
- Email is simple. It is easy to use, email allows for the easy and quick access of information and contacts.
- Email allows for easy referencing. Messages that have been sent and received can be stored, and searched through safely and easily.
- Email is accessible from anywhere – as long as you have an internet connection.
- Email is paperless, and therefore, beneficial for the planet.



- Email allows for mass sending of messages, you can send one particular message to several recipients all at once.
- Email allows for instant access of information and files.

DISADVANTAGES OF EMAIL

- Email could potentially cause information overload. Some messages may be dismissed or left unread, especially if there are a lot coming in and the network has not integrated some sort of email alert system into the computers at work.
- Email lacks a personal touch.
- Email can be disruptive. Going through each email can be disruptive to work as it does require a bit of time. This disruption is decreased through the utilisation of an email alert system.
- Email cannot be ignored for a long time.
- Emails can cause misunderstandings. Because email does not include nonverbal communication, recipients may misinterpret the sender's message. This is particularly true of senders fail to go through their messages before they send them.
- Email messages can contain viruses. It's best to be aware of this possibility so that you are careful when opening messages from people you don't know, or when downloading attachments.
- Email should be kept short and brief. This is especially difficult if you are one to send messages that are too long.
- Email requires timely responses. While some people tend to disregard messages, those that require responses should be replied to as soon as they are received and read. If not, urgent and important messages may be left untended.



Figure 7.2: Using email for communication purposes

When communicating by email, always note the following:

- Be polite.
- Do not write emails that are unnecessarily long.
- Do not spam people with emails that they are not interested in.
- Make sure that your message is logical and says what you intended to say.



Something to know

Asynchronous communication means that a message can be sent without the message receiver being available at the time. Most messaging applications are asynchronous. Most phone call applications, however, are synchronous.

SOME ADDITIONAL FEATURES

Email is not limited to only sending messages over the internet; it provides users with many features. We will now take a look at each of these in some more detail.

CALENDAR

Google Calendar is a time-management and scheduling calendar service developed by Google. It became available in beta release April 13, 2006, and in general release in July 2009, on the web and as mobile apps for the Android and iOS platforms.

Google Calendar allows users to create and edit events. Reminders can be enabled for events, with options available for type and time. Event locations can also be added, and other users can be invited to events. Users can enable or disable the visibility of special calendars, including Birthdays, where the app retrieves dates of births from Google contacts and displays birthday cards on a yearly basis, and Holidays, a country-specific calendar that displays dates of special occasions.

Over time, Google has added functionality that makes use of machine learning, including “Events from Gmail”, where event information from a user’s Gmail messages are automatically added to Google Calendar; “Reminders”, where users add to-do activities that can be automatically updated with new information; “Smart Suggestions”, where the app recommends titles, contacts, and locations when creating events; and “Goals”, where users enter information on a specified personal goal, and the app automatically schedules the activity at optimal times.

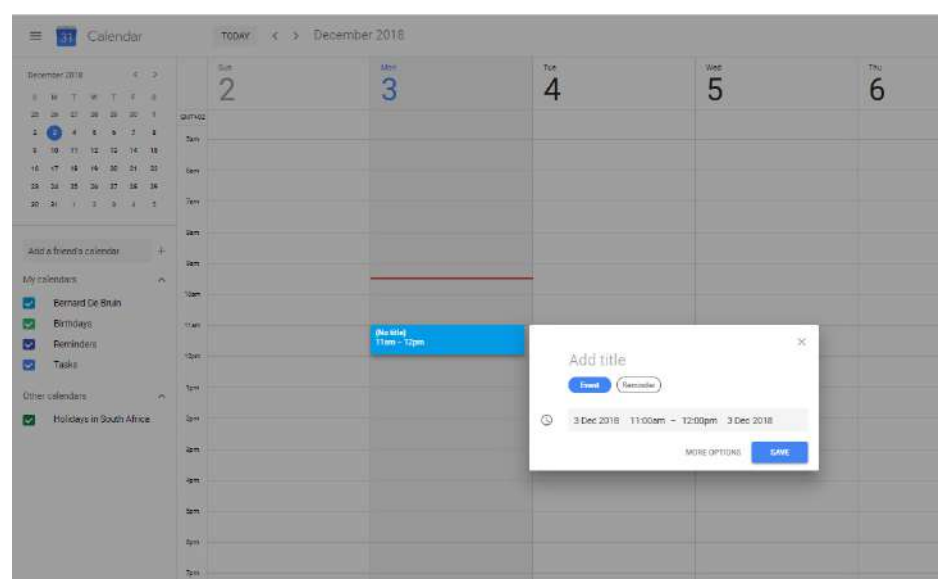


Figure 7.3: Set the date and time on Google Calendar

CONTACT LIST

You can now get to the contacts page by clicking the Apps icon in the upper right corner of the Gmail inbox. When you click the Apps icon, which is a square made up of nine smaller squares, it unfolds to reveal a panel of icons for other Google programs and services, including Google Photos, Google News and YouTube.



Figure 7.4: *Create new contact form*

TASKS

Gmail integrates a simple to-do list into your account. Google Tasks allows you to create lists of items, set due dates, and add notes. You can even create tasks directly from Gmail messages.

To add a task in in your Gmail account using Google Tasks, click the down arrow on the “Mail” menu in the upper-left corner of the Gmail window and select “Tasks.”

The “Tasks” window displays in the bottom-right corner of the Gmail window. Notice that the cursor is blinking in the first blank task. If the cursor is not blinking in the first empty task, move the mouse over it and click in it. Then type directly in the first blank task.

Once you’ve added a task, you can click the plus icon to create additional tasks. Pressing return after entering a task also creates a new task directly below it.

ARCHIVES

Rather than deleting an email and losing it for good, you can choose to archive it instead. As soon as a message is placed in the Gmail archive, it is removed from your inbox and tagged with the label All Mail. These messages remain in your Gmail account and can be retrieved easily at a later time, but in the meantime, they are out of sight and out of mind.

When someone replies to an archived message, it automatically returns to your inbox.

Sending a message to your Gmail archive is easy, so much so that some people mistakenly archive emails by clicking on or tapping the wrong option. Fortunately for them, retrieving archive messages is almost as easy as archiving them in the first place.

INSTANT MESSAGING

Instant messaging refers to short messages that are sent in real time over the internet. The messages can include multimedia items, such as pictures, videos and voice recordings.



ADVANTAGES OF INSTANT MESSAGING

- Messages are free to send
- Messages are received directly after being sent
- You can see if the message has been delivered
- You can see when your message has been read
- You can send a variety of messages; including text messages, pictures, videos, music and web links
- You can create group conversations in order to discuss a specific topic or plan events

DISADVANTAGES OF INSTANT MESSAGING

- Messages are not always saved
- It is an informal method of communication and might not be suited for business-related communications
- There is a pressure to respond immediately as people can see when you read their messages
- Can be distracting as one message can lead to a whole conversation
- Low security, as instant messaging services use a public network



Figure 7.5: Examples of instant messaging apps

When communicating by instant messaging, take note of the following:

- Do not expect an immediate reply. The person you are messaging might be busy and will reply once he or she is available.
- Keep your messages short and to the point.
- Do not type your messages using uppercase as it can be interpreted as shouting.
- Be polite.
- Do not use slang words and abbreviations. This might save you time, but it can also confuse others if they are not aware of the meaning.

VOIP

VoIP is a type of digital communication that allows the user to speak with one or more users over the internet. This type of communication is very similar to a phone call, with the exception being that it uses your internet connection and, therefore, uses data. Here are some of the advantages and disadvantages of VoIP.



ADVANTAGES OF VOIP

- VoIP is much cheaper than using traditional telecommunication services
- It can help with productivity as you can have face-to-face meetings with colleagues in different cities
- It saves time and money as you can have face-to-face business meetings without having to travel to the client

DISADVANTAGES OF VOIP

- You need an active internet connection
- Audio quality depends on the quality of your internet connection
- Some VoIP programs use large amounts of data

When communicating using VoIP, take note of the following:

- Be polite and speak clearly.
- Indicate that you are listening to the person with whom you are speaking.
- Repeat important details to ensure that you understand them correctly.
- Do not be afraid to ask questions if something is not clear.

VIDEO CONFERENCING

- Everyone can see you, all the time. This is not an audio conference, just because you are not speaking does not mean others in the conference can't still see you.
- Be punctual and courteous. Introduce yourself and take note of other attendees' names so you can address them by name. Turn off ringers for your other phones. Treat this just like you would an on-site meeting.
- No multi-tasking, we can see you. Look at your screen, pay attention to others and when speaking make sure to look at your camera.
- If it is improper for a face-to-face meeting, then it doesn't work for video either. Don't click your pen, tap on your desk or anything else annoying or distracting. Avoid yawning, gum chewing, etc.
- Make sure you have good light. Adjust lighting or use a portable light source to make sure you have good light shining at you from the front. You can overdo it too, so experiment until you find a good balance. Try pointing a strong desk lamp at the wall you're facing. You get good front light without having to look directly into a harsh light.
- Do not eat! You may enjoy a glass of water or coffee, but drinking a glass of wine or a bottle of beer is not acceptable. Do not smoke during the meetings either.
- Video allows us to do face-to-face meetings right from our virtual office. Even though we all enjoy sitting in a short and t-shirt in our virtual offices, it is not appropriate when you are called into a virtual meeting. Business casual at all times is the new rule. If you have a planned customer call you should consider dressing the same as you would for an on-site meeting.
- Do video calls from your desk or other appropriate location. Lying on the couch (or anywhere) with your pc on your chest or stomach doesn't present a flattering view.





- If you have a cluttered work space, make sure it's not showing up on camera. Consider removing award plaques from other (competitor) companies on your wall.
- Make sure to have current client version loaded before scheduled calls. Test your audio and/or video before a scheduled call.
- Avoid high traffic areas. Sometimes it's hard to avoid but do not position yourself where other people will be parading through the view of the camera on a regular basis.
- Close unused applications, video can be CPU/memory intensive.
- Avoid creating pixilation! Do not wear stripes, or anything with a heavy pattern. If you have vertical mini-blinds do not have them in the background. Minimise your hand gestures and body/head movements as well.
- Consider posting your comment/question in the chat window.
- Picture in Picture is your best reference, you can see yourself and your surroundings just as others on the call can. Pay close attention to what you see there, and make adjustments as necessary.
- DO NOT video while driving.



Activity 7.1

1. Write down the correct answer for each of the following questions.
 - a. Which of the following is NOT an additional email feature?
 - A. Contact list
 - B. Calendar
 - C. Tasks
 - D. Voice calls
 - b. Which of the following is NOT a type of electronic communication?
 - A. Instant messaging
 - B. Posting a letter
 - C. Email
 - D. Video conferencing
 - c. Which of the following is the most common netiquette when using electronic communication?
 - A. Be polite
 - B. Be there
 - C. Mute everyone
 - D. Explain everything thoroughly.





Activity 7.1

... continued

2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Allows users to save their emails for later use	A. Video conference
2.2 The advantage of being able to save messages for future reference	B. Archiving
2.3 Low security is a disadvantage since a public network is being used	C. Tasks
2.4 A direct conversation with multiple people at the same time	D. Email
2.5 Much cheaper than traditional telecommunication services, but you need an active internet connection	E. Calendar
	F. Netiquette
	G. Instant messaging
	H. VoIP

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if it is false.
- Do not say things to people you would not say to them in real life.
 - On the internet, there is no need to show people respect.
 - When joining an existing conversation, speak about the subjects relevant to you.
 - Make sure your messages are clearly written and easy to understand.
 - Things posted on the internet do not last forever. This means that things you post as a teenager or young adult will not affect the rest of your life.
4. Answer the following questions:
- Explain the world wide web to your peer.
 - List the disadvantages of electronic communication.
 - Explain the advantages of email.
 - Define 'netiquette' and describe proper netiquette when communicating on email.
4. VoIP software, such as Skype, is becoming increasingly popular to use.
- Give TWO reasons why more people are currently using Skype than in the past.
 - Give ONE reason why the video and sound quality is sometimes poor when using Skype to make a video call.
 - Suggest ONE way in which the sound quality of a VoIP call can be improved without ending the session or changing any hardware.



7.2 Electronic communication

Web browsers are software applications that allow users to access information on the world-wide web. This is done by allowing users to open and view web pages containing images, videos, text, as well as links to other web pages. The most common web browsers include Microsoft Edge, Google Chrome, Safari and Firefox.

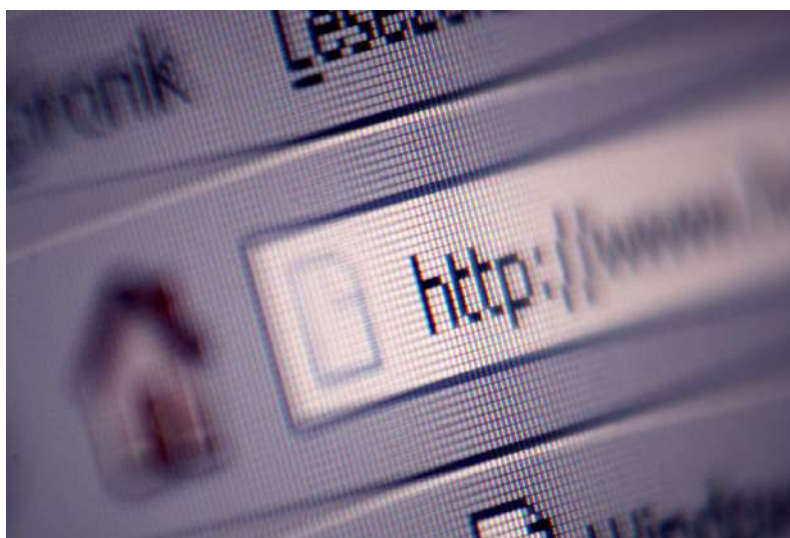


Figure 7.6: Types of web browsers

In the following section, we will look at some of the features that have been included in web browsers to help make your internet browsing experience as safe and as enjoyable as possible.

Something to know

Some web browsers, for example Internet Explorer, use the term “Favourites” instead of “Bookmarks”. Do not be alarmed by this, as they perform the same function.

BOOKMARKS

When you think of a bookmark, you most likely think of a piece of paper or card that you place in a book in order to mark the page you are currently reading. This same principle applies to the bookmarks used in your internet browsers. A bookmark is a saved shortcut that tells your browser to go to a specific website address. This saves the user the effort of having to remember and type the website address or URL each time he or she visits the page.

Depending on your web browser, you can also bookmark a web page by clicking on the star icon to the right of the website URL. This will automatically add the address to your bookmarks.

To add a new bookmark in Firefox 4, the latest version of Mozilla’s Web browser, click on the star-shaped icon next to the website URL, located in the address bar at the top of the browser window. This adds the website currently opened to the list of bookmarks. To view bookmarks, click on the book-shaped icon with the star on it, located on the top right of the browser window.

bookmarks, click on the wrench-shaped icon at the top right of the browser window (next to the star-shaped icon). Select “Bookmark Manager” from the drop-down menu.

To add a bookmark in Apple Safari, click the “Bookmarks” menu at the top of the browser window, then select “Add Bookmark” from the drop-down menu. Click the “Add” button. This adds the website you’re currently browsing to your bookmarks. To view a list of saved bookmarks, click the “Bookmarks” menu at the top of the browser window and select “Show All Bookmarks.”

To add a bookmark in Google Chrome, click on the star-shaped icon located in the address bar at the top of the browser window. A dialogue box appears to confirm the new bookmark. Click “Done.” This adds the website currently opened to the list of bookmarks. To view

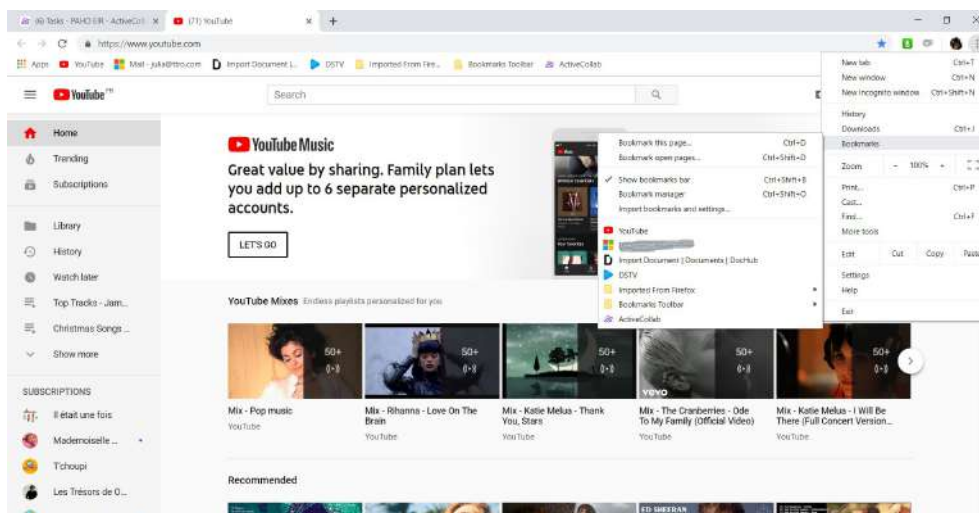


Figure 7.7: Creating and accessing bookmarks in Google Chrome

HISTORY AND FAVOURITES

History list organizes the list of webpages that are visited over a period of time by category. History List contains the URLs for the websites and page that visited through internet explorer browser. The history will save the list of webpage URL’s by Today, Yesterday, Two weeks ago and Three weeks age and so on to easily find the date and URL of the visited page on that particular date.

Favourites saves them like if you click add to favourites in the options with the particular webpage open; then it adds the website you are currently browsing to your favourites list.

IMPORTING FAVOURITES AND BOOKMARKS

Import your favourites or bookmarks from one browser to another if you’re switching browsers or would like to use more than one. To import bookmarks or favourites into Firefox 4, open Firefox and press the “Alt” key to temporarily bring up the “File” menu. Click the “File” menu and select “Import.” An “Import Wizard” is then displayed. Select the browser from which you wish to import bookmarks or favourites and click “Next.” Select which items to import and click “Next.”



Something to know

Most browsers contain a section called *Recently closed*. This section forms part of your browser history and is a list of the web pages that you recently viewed and closed.



Something to know

To access the *History* function in Google Chrome:

- Click the *Open* menu icon.
- Go down the menu and click *History*. This should open a new window containing your web-browsing history.
- Scroll down the list of websites and select the one you are looking for.



Something to know

To activate the ad blocker in Google Chrome:

- Click the *Menu* icon.
- Go down the menu and click *Settings*. This should open a new tab in your web browser.
- Scroll to the *Privacy and security* options and click *Control Settings*. This will take you to a new menu.
- Scroll down the menu and click *Pop-ups and redirects*.
- Select the *Blocked (recommended)* checkbox. You can now close the tab, as the settings are automatically saved..



Something to know

Websites can also be blocked by changing the settings on your router, using specific software and through Windows. If you want to block a website using Google Chrome, you can download a browser plug-in from the Chrome web store.

To import bookmarks as favourites in Internet Explorer 9, click on the star-shaped icon on the top right of the browser window. Click on the down-pointing arrow next to “Add to favourites” and select “Import and Export.” Select “Import from another Browser” and click “Next.” Tick the box next to the name of the browser from which you wish to import bookmarks and click “Import.”

HOME PAGE SETTINGS

A home page is a webpage that serves as the starting point of website. It is the default webpage that loads when you visit a web address that only contains a domain name. For example, visiting <https://techterms.com> will display the Tech Terms home page.

SET YOUR HOME PAGE

You can customize Google Chrome to open any page for the homepage or startup page. These two pages aren't the same unless you set them to be.

- Your start up page is the one that shows when you first launch Chrome on your computer.
- Your homepage is the one you go to when you click Home .

CHOOSE YOUR HOMEPAGE

You can control what page appears when you click Home.

- On your computer, open Chrome.
- At the top right, click More and then Settings.
- Under “Appearance,” turn on Show Home button.
- Below “Show Home button,” choose to use the New Tab page or a custom page
- The Home button will appear to the left of your address bar.

BLOCKING WEBSITES

Website Blocking is a process by which a Firewall or WW Proxy prevents users from accessing some network resources, such as Word Wide Web sites or FTP servers.

Site Blocking is a legally dangerous practice, since it prevents the organisation that employs it to effectively argue that their Internet connection constitutes a Common carrier. This may make the organisation liable for offensive or illegal documents downloaded from the Internet.

HOW TO BLOCK WEB SITES IN GOOGLE CHROME

Do you find yourself accessing Facebook, Twitter, or other distracting websites when you really should be doing something else instead? Block that website in Google Chrome. Other solutions will let you block websites for children at home or employees, too.

We recommend Block Site for Chrome for quickly blocking websites. Install it, and you can use the extension's simple options to define a list of blocked sites. You can even set up a redirection, so you're automatically pointed at a website you should be using (perhaps your





workplace's website) when you visit a blocked site. Or, enable it on a schedule so you can look at Facebook all you want—as long as it's outside of the hours you should define.

This isn't fool proof. In fact, it's the opposite. You can quickly edit Block Site's permissions to unblock websites. And, while you can define password protection, anyone using your computer could open any other web browser to get around the block. It's a way to keep yourself on track and add some additional friction before you access a website—that's all. It's not a way to control access to websites for children at home or employees in an organisation

If you don't want an extension like Block Site to run on every website you visit, you can use Google Chrome's extensions permissions to restrict its access. For example, if you just want to block Facebook, you could restrict Block Site to only run on facebook.com.

CACHING

Caching is the temporary storage of web documents such as HTML pages and images. Basically your web browser stores copies of web pages you've visited recently to reduce its bandwidth usage, server load, and lag.

This could be why our website is offline, why it seems like your name servers still aren't set, or why you can't see the changes you've made to your website.

If you suspect this is the case, you can first try clicking “refresh” on your browser a few times. If this still does not work, you can clear your browser's cache to see if that is the problem. Each browser is different, but here are some links to the most popular browsers cache clearing instructions:

- Chrome
- Firefox
- Safari
- Internet Explorer
- Opera

Another good tip is to visit the website on a device after turning OFF the wifi. This should get you a real-time view page and confirm a local caching issue on the original machine.

BROWSER PLUG-INS

A plugin is a piece of software that acts as an add-on to a web browser and gives the browser additional functionality. Plugins can allow a web browser to display additional content it was not originally designed to display.

An example of a plugin is the free Macromedia Flash Player, a plugin that allows the web browser to display animations using the Flash format. As the Web has become more commercial, Flash has become a popular format for displaying ads in web pages. As a result, many web users have been prompted to download the Flash plug in and have it installed on their systems.



Something to know

If you are using a Windows operating system, then the folder used for the web cache is known as *Temporary Internet Files*.





Other popular plugins include Quicktime Player (available on the Apple website) and Acrobat Reader (which in addition to being a plugin for the major browsers is also a stand-alone application used to display files using the PDF format).

Most plugins are available as free downloads. To install the plugin, you visit the website of the plugin's developer and click on a link that will download the installer for the plugin you have selected. You can save the installer to an easy to find location such as the Desktop or a specific folder you have created to organize all of your downloads. Once you have downloaded the installer, you can open it and follow the prompts to install the plugin on your system. You may have to restart your web browser to enable the additional functionality provided by the plugin.

POP UP BLOCKERS

Pop-ups are frequently used by websites to advertise products or features. They are therefore meant to be as eye-catching as possible. This is achieved by making the pop-up open in a small secondary window, which becomes the active window. To get the maximum amount of attention, a pop-up may also feature bright colours, animation and motion. For browsers that support tabbed browsing, the pop-up may open in a secondary tab instead of a whole new window. Pop-ups are commonly generated using JavaScript.

Pop-up blocking is usually enabled by means of a check-box that one must tick to enable pop-up blocking, or untick to disable. All the major browsers now support pop-up blocking.

Pop-up blockers may also be installed as third-party software tools. These typically incorporate additional features such as ad filtering and highly customizable pop-up blocking options. For the majority of users, however, the blockers built into all major browsers are perfectly adequate.

Not all pop-ups are a nuisance. In fact, some are very helpful. For example, pop-ups are often used to provide guidance to users on how to fill in a form on a Web page. Unfortunately, modern browsers may accidentally block these as well (hence the ability to enable or disable the whole feature). Some browsers have the ability to discern which pop-ups are genuine tools on the website and treat them differently from pop-up advertisements, a feature called intelligent blocking. Some browsers indicate to the user when they have blocked a pop-up, usually by means of a small information bar lasting a few seconds, an audible signal, or both.

Example 7.1 How to view all the installed plug-ins in Google Chrome

1. Click the *Open* menu icon.
2. Click *Settings*.
3. Click *Advanced*.
4. Click *Content Settings*. This will open a new menu.
5. Scroll down and select the plug-in you would like to view.



PRIVATE BROWSING

Since 2008, January 28th has been set aside for Data Privacy Day. The goal: “to create awareness about the importance of privacy and protecting personal information.”

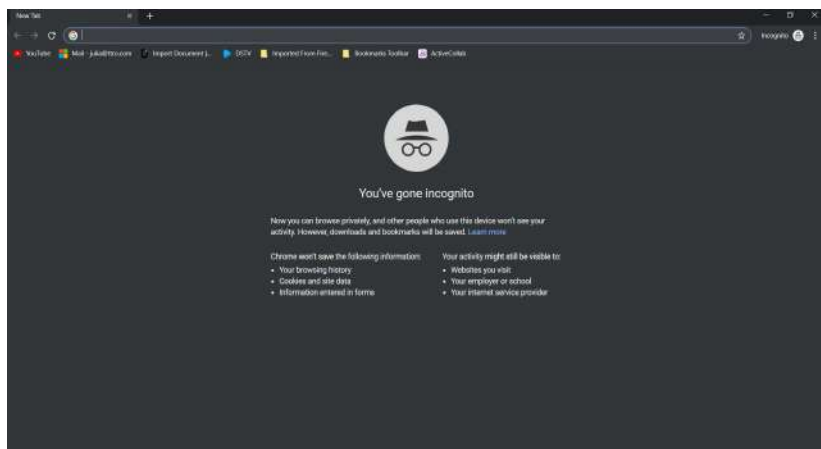


Figure 7.8: People use private browsing for a variety of reasons

It's the perfect time to take a look at one privacy feature that's right in front of you: your web browser's private browsing mode. Just what is it that makes private browsing private? Let's take a look at the major browsers and see.

GOOGLE CHROME

Google Chrome calls it *Incognito Mode*, and you can tell you're using it by looking for the “secret agent” icon in the top left corner of the window. Chrome also shows you a big, bold new tab page when you open an Incognito window. That's it at the top of this post. In *Incognito Mode*, Chrome won't keep track of the pages you visit, the data you enter into forms, or any searches you submit. It won't remember what files you download, but those files will stay on your computer after you close the Incognito window. You'll have to manually delete them if you want them gone. The same goes for bookmarks you create.

Example 7.2 How to use private browsing with Google Chrome

1. Click on the *Open* menu icon.
2. Click on *New incognito* tab. This will open a new private browser.

You can also open a private browser by pressing *Ctrl+Shift+N*.

Activity 7.2

1. To create bookmarks, you can do the following:
 - a. Open your Google Chrome browser.
 - b. Create four bookmarks.
 - c. Pin your bookmarks on the browser.
1. A variety of plug-ins can be installed on web browsers.
 - a. Why would you use or need browser plug-ins?
 - b. Give ONE common example of a browser plug-in.

 Activity 7.3

1. Which is NOT an example of a web browser?
 - a. Avast
 - b. Microsoft Edge
 - c. Google Chrome
 - d. Firefox
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Another term used for bookmarks	A. Pop-up ad
2.2 A small area allocated on your hard drive in order to temporarily store images and pages from the world-wide web	B. Bookmark
2.3 What a star icon on a web browser indicates	C. Her story
2.4 Documenting the websites that you previously visited	D. Favourites
2.5 A digital advertisement that appears when you open a specific web page	E. Web cache
	F. History
	G. Disk cache

3. Define a bookmark.
4. State TWO disadvantages of adware, other than the fact that it might be annoying.

In the first unit of this chapter, we discussed different types of digital communication, as well as their advantages and disadvantages. In this section, we will be taking a closer look at how these different types of digital communication can be used in order to communicate with others.

BLOGGING

A **blog** (shortening of “weblog”) is an online journal or informational website displaying information in the reverse chronological order, with the latest posts appearing first. It is a platform where a writer or even a group of writers share their views on an individual subject.

ADVANTAGES OF BLOGGING

- Enables you to write down your thoughts on anything of interest
- Very quick and easy to set up
- Easy and quick to update and add a new post
- People can leave comments on your blog

DISADVANTAGES OF BLOGGING

- Whatever you publish is available for everyone to see.
- Personal blogs may be biased
- Blogs can be time consuming
- People may leave rude and inappropriate comments

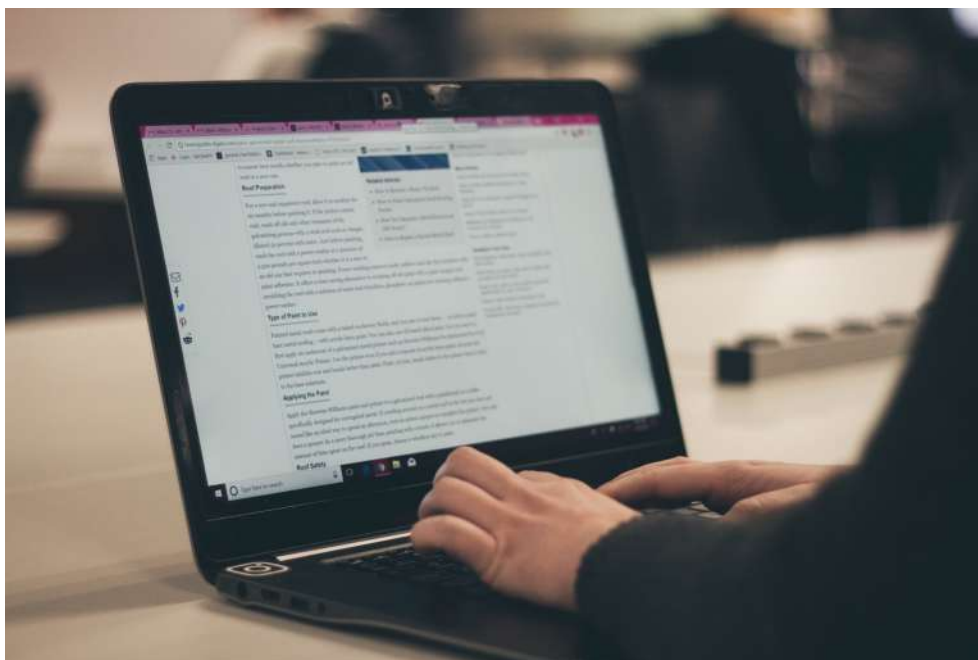


Figure 7.9: *Blogging as a form of communication*

CREATING A BLOG, REGISTERING A BLOG SPACE AND PUBLISHING A BLOG

The easiest way to create a blog is to register a blog space using a free blogging service, such as WordPress or Blogger.com.

This not only enables you to create a blog with little effort; it also makes it easier for readers to find your blog as it is associated with the blogging platform.

The first step in creating your blog is to register a blogging space. This can be done in the following way:

1. Make sure you have an internet connection and a Gmail account. (Some services do not require a Gmail account, but do require an email account.) If you do not have a Gmail account, make sure to register a Google account. (The Google account includes a Gmail account.)
2. Go to the blogging service website by typing the URL into your browser, or by searching for the service in Google.

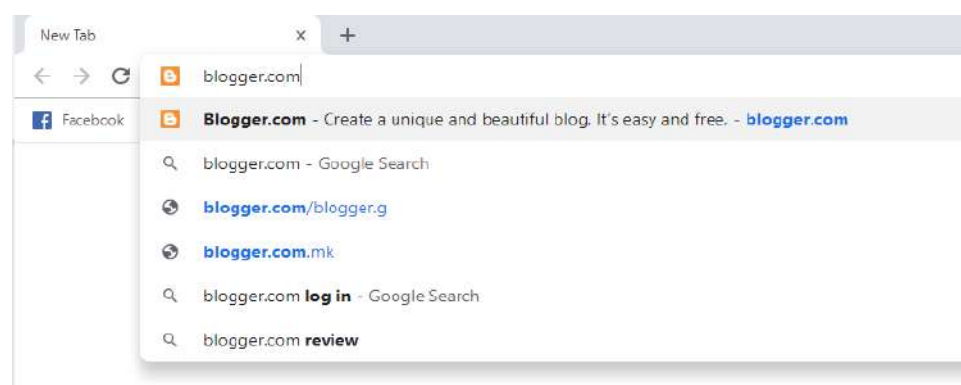


Figure 7.10: Registering a blog space

3. Once you are on the blogging service website, click *Get Started*, or *Create Your Blog* to register a new account. (The option will differ, depending on the website.)
4. Complete the account registration process by following the step-by-step instructions. This will include choosing an address for your blog.

Once you have registered your blogging account, you are ready to start creating your blog by writing your first post.

To create posts for your blog by using a word processor, such as Microsoft Word, you can do the following:

1. Open Microsoft Word.
2. Click *File* and go to *New*.

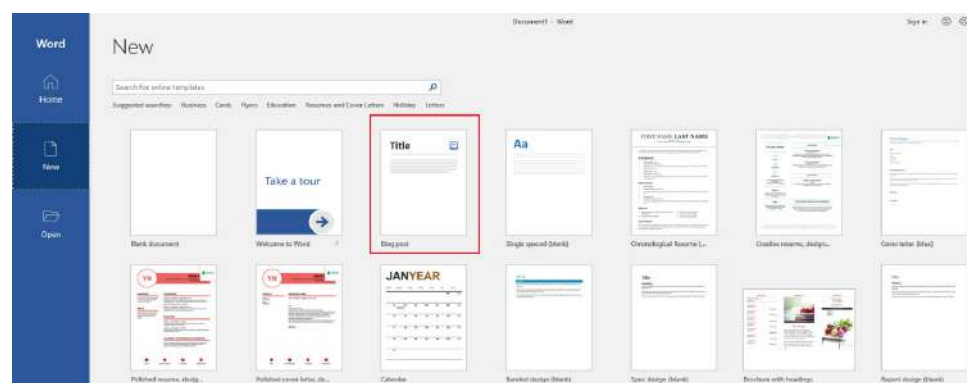


Figure 7.11: Creating a blog using a word processor

Something to know

Users will be able to find your blog by typing the address that you registered into their browsers.

3. Select *Blog post* from the list of new documents. This will open a preview of the blog post.



Figure 7.12: A preview of the blog post

4. Click *Create*. This will open the *Register a Blog Account* window.

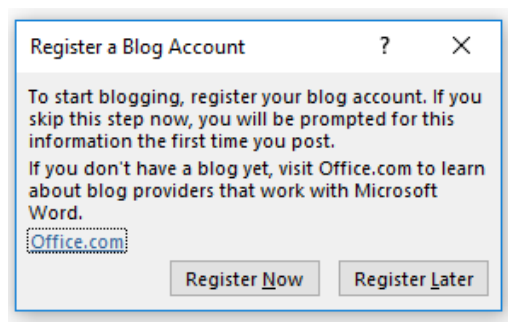


Figure 7.13: Registering a blog account

5. Click *Register Now*.
6. Select the service where you have registered a domain name from the blog drop-down list and click *Next*.

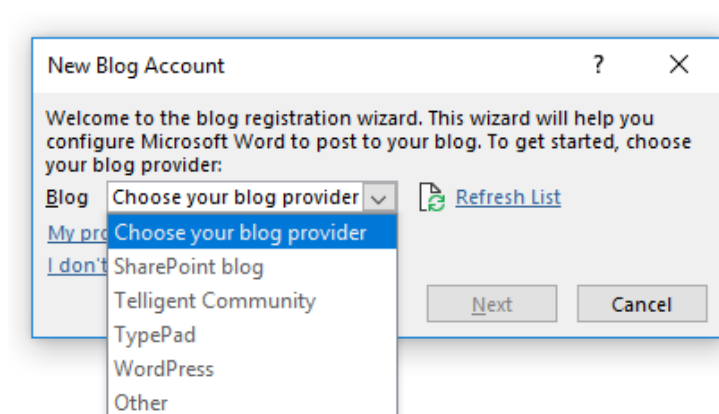


Figure 7.14: Choosing the blog provider

7. Fill in the details of your domain and click *OK*. This will allow you to start creating your blog in Word.

Something to know

You do not need to use a blogging service in order to create a blog. You can also register your own domain name and use it to host your blog. This method has some advantages, such as allowing you the freedom to move your blog to a new domain while keeping all your readers. However, it should be noted that there will be some costs involved in hosting your own website.

Figure 7.15: Filling in the details of the domain

8. Once you have created content for your blog, click *Publish*.

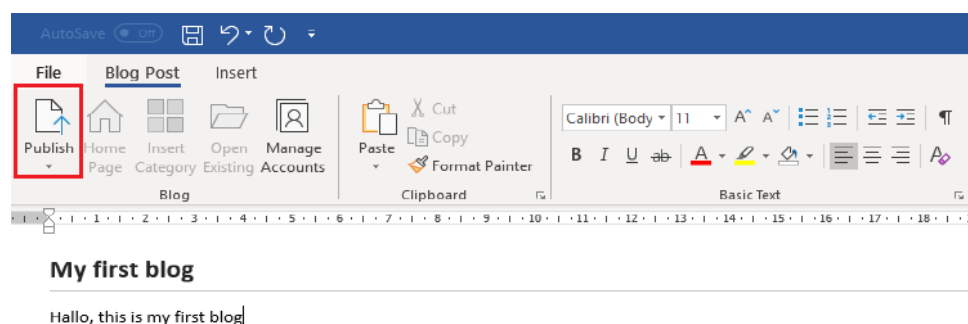


Figure 7.16: Publishing a blog

VLOGGING

A video blog or video log, usually shortened to **vlog**, is a form of blog for which the medium is video, and is a form of web television. Vlog entries often combine embedded video (or a video link) with supporting text, images, and other metadata. A vlog is like a personal website or social media account where a person regularly posts short videos. Entries can be recorded in one take or cut into multiple parts.

ADVANTAGES OF VLOGGING

- Easy to connect with an audience
- You do not have to create your own website
- Does not need to be professional

DISADVANTAGES OF VLOGGING

- Requires basic recording equipment
- Requires basic video editing knowledge
- Can be very time consuming



PODCASTING AND VODCASTING

Podcasting is a free service that allows internet users to pull audio files (typically MP3s) from a podcasting website to listen to on their computers or personal digital audio players. The term comes from a combination of the words iPod (a personal digital audio player made by Apple) and broadcasting. A podcast is an audio programme produced on a regular basis, delivered over the internet in a compressed digital format and designed for playback on computers or portable digital audio players, such as the iPod.

A **vodcast** is a podcast consisting of video recordings, instead of solely audio.

ADVANTAGES OF PODCASTING AND VODCASTING

- Good to listen to while travelling
- A good way to get a daily update
- Can be professional or relaxed
- Can be very informative

DISADVANTAGES OF PODCASTING AND VODCASTING

- Can be time consuming and difficult to make, especially scripted podcasts
- Needs a good reader, good information and good writing in order for it to be interesting
- Audience for podcasts is smaller

BEST PRACTICE

Podcasts or vodcasts can cover a variety of subjects; including novels, dramas, news, or even talk shows discussing daily events or the latest games. To help you start your very own podcast or vodcast, you can do the following:

- Ensure that you have enough time to script, record and edit the podcast or vodcast.
- Stay consistent with your uploads; schedule your updates so that listeners will know when new content will be posted.
- Be polite, and speak naturally and clearly.
- Choose topics about which you are passionate.
- Advertise your podcast or vodcast. Sharing the release and topic information on social media is a good way in which to obtain new listeners.

WIKIS

Wiki websites are a form of digital communication that are used to obtain and share information with other users. The website can be accessed and edited by any user across the world, so long as that person has a computer and an internet connection. Due to this reason, it is very important to verify any information obtained from a wiki.

ADVANTAGES OF WIKIS

- Anyone can edit the information and, therefore, is updated regularly
- Easy and free to use
- Can be accessed from any device that has an internet connection





DISADVANTAGES OF WIKIS

- Anyone can edit the information, which means it can lead to misinformation
- Open to spam and vandalism if not moderated
- Needs an active internet connection

The most famous wiki website is Wikipedia, an online encyclopaedia. However, there are many other wiki websites; each covering a specific topic.

RSS FEEDS

WHAT IS RSS?

RSS stands for “Really Simple Syndication”. It is a way to easily distribute a list of headlines, update notices, and sometimes content to a wide number of people. It is used by computer programs that organize those headlines and notices for easy reading.

WHAT PROBLEM DOES RSS SOLVE?

Most people are interested in many websites whose content changes on an unpredictable schedule. Examples of such websites are news sites, community and religious organization information pages, product information pages, medical websites, and weblogs. Repeatedly checking each website to see if there is any new content can be very tedious.

Email notification of changes was an early solution to this problem. Unfortunately, when you receive email notifications from multiple websites they are usually disorganized and can get overwhelming, and are often mistaken for spam.

RSS is a better way to be notified of new and changed content. Notifications of changes to multiple websites are handled easily, and the results are presented to you well organized and distinct from email.

HOW DOES RSS WORK?

RSS works by having the website author maintain a list of notifications on their website in a standard way. This list of notifications is called an “RSS Feed”. People who are interested in finding out the latest headlines or changes can check this list. Special computer programs called “RSS aggregators” have been developed that automatically access the RSS feeds of websites you care about on your behalf and organize the results for you. (RSS feeds and aggregators are also sometimes called “RSS Channels” and “RSS Readers”.)

Producing an RSS feed is very simple and hundreds of thousands of websites now provide this feature, including major news organizations like the New York Times, the BBC, and Reuters, as well as many weblogs.

RSS FEED PROVIDE THE FOLLOWING INFORMATION

RSS provides very basic information to do its notification. It is made up of a list of items presented in order from newest to oldest. Each item usually consists of a simple title describing the item along with a more complete description and a link to a web page with





the actual information being described. Sometimes this description is the full information you want to read (such as the content of a weblog post) and sometimes it is just a summary.

GPS

GPS is a form of digital communication that uses satellite-based radio navigation in order to determine where you are in the world. This information can then be used to navigate to a new destination, or to let others know where you currently are.

ADVANTAGES OF GPS

- GPS is available from anywhere in the world
- GPS can be used to locate a person quickly and efficiently if he or she is in trouble
- GPS software is updated regularly to account for any change to the landscape
- GPS can be used to plan your route, get directions and find retail outlets

DISADVANTAGES OF GPS

- GPS devices use a lot of power, which requires that the battery be charged regularly
- GPS signal might be affected by weather, tall buildings and structures
- GPS accuracy depends of the strength of the signal; a weak signal can lead to inaccurate information, which is why we recommend that you always bring a map as backup
- GPS devices are expensive to purchase

In the past, users had to buy a GPS device, such as a TomTom, in order to receive and analyse signals from a satellite network. This is no longer the case. Modern smartphones come equipped with a built-in GPS, as well as navigating software, such as Google Maps.



Figure 7.17: *Using GPS to navigate your way around*





Something to know

Due to their ability to connect people, many of the world's largest websites are social networking sites. This includes Facebook, Twitter and Instagram.

GEOTAGGING

Geotagging is the process of attaching geographical data (obtained using a GPS) to media and websites. This includes attaching latitude and longitude coordinates, timestamps, altitude, distance and place names to videos, photos, messages and QR codes.



Figure 7.18: Geotagging is used to attach geographical data to media and websites

SOCIAL NETWORKS

A social networking service is an online platform that allows the user to communicate and share information with other users from around the world. In simple terms, it is a website that connects you to people by making friends and seeing what they are up to. You can usually also post pictures and share photos, videos and ideas with an individual (through direct message) or group of friends (through posting a message).

ADVANTAGES AND DISADVANTAGES OF SOCIAL NETWORKS

- Helps the user to stay connected with friends and family
- Helps the user to meet people with common interest
- Can be used to promote a product and find products in which you are interested
- Lets the user stay up to date with the latest news from around the world

ADVANTAGES AND DISADVANTAGES OF SOCIAL NETWORKS

- Cyberbullying is a growing problem in social media
- Information obtained from social media can be used to profile and discriminate against people
- Some people can get addicted to using and posting on social media
- Social media is a big source of false and misleading information



Figure 7.19: Social media sites



CREATING A NEW FACEBOOK ACCOUNT

You can create a social media account on a site such as Facebook, by going to www.facebook.com and completing the *Create an account* form. Once you are done, click *Sign Up*.

Figure 7.20: Creating a new Facebook account



Activity 7.4

1. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
1.1 A standardised format that is used to summarise stories or articles that can be found on a website	A. GPS
1.2 A form of digital communication that uses satellite-based radio navigation in order to determine where you are in the world	B. Vodcasting
1.3 Similar to blogging; with the major difference being that it uses video instead of written stories and articles	C. RSS feeds
1.4 Type of digital communication that uses an audio-only broadcast	D. Best practice
1.5 Uses both audio and video for the broadcast	E. Wikis
	F. Vlogging
	G. Podcasting
	H. Geotagging

... continued



Activity 7.4

... continued

2. Christy wants to create a website to share her make-up tips with others. Do some research and write a short essay including the following information:
 - a. Advise Christy on what type of computer communication she should use.
 - b. Inform Christy of five advantages of this type of computer communication.
 - c. Inform Christy of five disadvantages of this type of computer communication.
3. Answer the following questions:
 - a. List three advantages and one disadvantage of blogging.
 - b. Explain how you would create and publish a blog using Microsoft Word.
 - c. Give one disadvantage of wiki websites.
 - d. Explain what geotagging is.
 - e. Explain how cyberbullying works and give an opinion on how to prevent cyberbullying, without taking away the technology.





UNIT

7.4 Communication devices

Throughout this chapter, we have taken a look at electronic communication, what it is, the different types, as well as its uses. In this section, we will be investigating some of the devices that you can use to communicate with your friends, family, colleagues, or even strangers over the internet.

COMPUTERS AND LAPTOPS

Desktops and laptop computers have become a big part of our everyday lives. They enable us to make use of a variety of different digital communication methods, including the following:

- Written communication, such as instant messaging and email
- Audio communication, such as VoIP
- Video communication, such as video conferencing

Computers can also be found in a variety of electronic devices where it enables the system to communicate with itself in order to function properly.

SMARTPHONES

The smartphone is a type of computer that people can carry in their pockets and is thousands of times more powerful than the computers that were used to put the first man on the moon. The smartphone can connect to the internet and can perform a variety of different functions, including the following:

- Making and receiving calls
- Sending and receiving written messages
- Sending and receiving audio messages
- Video conferencing
- Social networking
- GPS
- Listen to music
- Take photos and videos
- Play a variety of mobile games



WEARABLES

Wearables are a new category of computers, which consist of small devices that have been designed to be worn, or attached to your body. Examples of these include the following:

- **Smartwatches:** Smartwatches can be synced with your smartphone, allowing you to see your messages and calls on your watch.
- **Smart clothing:** Smart clothing is specially designed clothing to monitor a user's behaviour, such as the heart rate and blood glucose levels.
- **Fitness trackers:** Fitness trackers allow users to record their daily activities, workouts, heart rate, number of steps and number of calories burned. These trackers are promoted by healthcare companies.

TABLETS

Tablets are small hand-held computers that are smaller than a laptop, but bigger than a smartphone. Modern tablets work in a way very similar to that of smartphones. The main differences between the two are that tablets are bigger and are normally used for different activities. For example:

- Tablets are traditionally used by the whole family to watch shows at home
- Smartphones are used when the owner of the phone leaves home to communicate with others



Activity 8.5

1. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
1.1 A type of computer that users can carry in their pockets	A. Wearables
1.2 Computers that consist of small devices, which have been designed to be worn or attached to the body	B. Smartphone
1.3 Small hand-held computers that are smaller than a laptop, but bigger than a smartphone	C. Tables
1.4 Hand-held digital device that only has the following functions: calendar, calculator, notepad and appointment book	D. Tablets
1.5 An example of blockchain technology	E. 5G

2. Answer the following questions:
 - a. What is the difference between a smartphone and a tablet?
 - b. Name and describe four functions of a smartphone.
 - c. Describe the three different digital communication methods that enable global connection.



REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

- 1.1 Which of the following is a disadvantage of electronic communication? (1)
- A. Fast and easy to use.
 - B. Can be used for a wide variety of applications.
 - C. People can become addicted to certain forms of digital communication.
 - D. Saves you money as it is cheaper than traditional communication methods.
- 1.2 Which of the following is NOT proper netiquette? (1)
- A. Show people on the internet the same respect you would show to them in real life.
 - B. Make sure your messages are clearly written and easy to understand.
 - C. When joining an existing conversation, speak about subjects that interest you.
 - D. Try to make useful contributions and help people out on the internet.
- 1.3 Which of the following uses asynchronous communication? (1)
- A. Email
 - B. Instant messaging
 - C. Video conferencing
 - D. VoIP
- 1.4 Which web browser comes preinstalled on Windows XP? (1)
- A. Microsoft Edge
 - B. Internet Explorer
 - C. Google Chrome
 - D. Safari
- 1.5 Which of the following can be used to prevent a person from accessing dangerous content on the internet? (1)
- A. Bookmarks
 - B. Private browsing
 - C. Website blockers
 - D. Caching

[5]

QUESTION 2: TRUE OR FALSE

Indicate if the following statements are TRUE or FALSE. Choose the answer and write true or false next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word 'NOT' to change the statement.)

- 2.1 It is proper etiquette to be careful about the things you say to people online. (1)
- 2.2 When you write instant messages you must not use slang and symbols. (1)
- 2.3 VoIP requires more bandwidth than instant messaging. (1)
- 2.4 During VoIP mute yourself when you are not speaking. (1)
- 2.5 You can import bookmarks from one web browser to another. (1)

[5]



REVISION ACTIVITY

... continued

QUESTION 3: MEDIUM QUESTIONS

- 3.1** Which of the following types of electronic communication describes the disadvantages in the table below the best? (6)

COMMUNICATION TECHNIQUES
Blogs
Wikis
VoIP
IM
Video conferencing

DISADVANTAGE	COMMUNICATION TECHNIQUES
a. Anyone can edit the information which can lead to misinformation.	
b. It is an informal method of communication and might not be suited for business related communications.	
c. It uses a significant amount of data which can be expensive.	
d. Can be time consuming to keep updated.	
e. Designed for letter writing, rather than having conversations.	
f. You need an active internet connection.	

- 3.2** Name two web browser features that you can use daily. (2)
3.3 Describe three different digital communication methods that enables you to connect globally. (3)
3.4 What is geotagging? Give an example of when you would make use of geotagging. (2)
3.5 Briefly explain how companies can benefit from knowing your search history. (2)

[15]

TOTAL: 25

AT THE END OF THE CHAPTER

NO.	CAN YOU ...	YES	NO
1.	Explain what digital communication is?		
2.	Identify and discuss the different types of digital communication?		
3.	Describe the most common features of a web browser?		
4.	Discuss some of the most popular communication devices?		
5.	List and describe the uses of computer communications?		



Unit 8.2 Impact of technology



- Discuss the impact that social media has had on the community.
- Identify and list the benefits and harm of technology.
- Explain what information overload is.
- Describe the impact of virtual reality.

The fastest way to advertise anything is to upload it on social sites. Social media is faster in conveying the news or information than any other media say radio, television or newspapers. Also, government sites as well as private sites are available for filing documents. Ticket booking has become very serviceable to the customers, may it be for a movie show, a hotel in a far town or for an overseas travel trip.

Through many years, social media has helped people stay in contact with their distant friends, relatives and other influential people in their lives, which would have otherwise been impossible. Hence social media proves to be an excellent platform to promote and facilitate innumerable things.



Figure 8.1: *We rely heavily on technology in modern times*

8.1 Social networking sites and technologies

Modern technology and the internet have made it easier than ever to meet new people by using social networking sites. A social networking website is an online platform that allows users to create a digital profile of themselves; including their likes, dislikes, personal information and much more. This profile is then used to build a network of friends with whom the user can communicate and share information.

- **Facebook:** This is the most popular social networking site in the world. Facebook allows the user to create a profile that can be used to upload photos and videos, send messages, and keep in touch with friends and family.
- **Twitter:** Twitter is a social media and news platform that allows users to share their thoughts, ideas and opinions with other users, in 280 characters or less.
- **Instagram:** Instagram is a social network that allows users to share photos and videos with other users.
- **YouTube:** YouTube is a video-sharing platform that allows users to upload and view videos over the internet.
- **GoFundMe:** GoFundMe is a social networking site that allows users to donate money (crowdfund) towards a cause or project.
- **WhatsApp:** WhatsApp is a messaging app for smartphones where messages, images, audio and video can be sent using the internet.
- **LinkedIn:** LinkedIn is a professional networking site that is used to display your business or professional profile. It can be used to apply for jobs.



Figure 8.2: Examples of social networking sites



IMPACT OF SOCIAL NETWORKING SITES

There can be little doubt that social networking sites have changed the way in which we interact with each other. From Facebook to Instagram, social networking sites allow users to find and connect with people from around the world, based on your personal information and interests. But how safe is that information?

POSITIVE EFFECTS OF SOCIAL MEDIA

Social media has changed the way in which we make friends, talk to each other and obtain news. Let's take a look at some of the additional positive effects of social media:

- **It can be used for education:** Over the last few years, YouTube has become a platform where people can find educational videos on a variety of subjects. This can include something as simple as how to cook a steak; to more complicated subjects, such as Mathematics and Science.
- **It can be used to raise awareness:** Sites like Facebook and Twitter have made it easier than ever for people to raise awareness around a special cause. This can include creating a support group for people suffering from cancer; to raising awareness of how to treat people with disabilities.
- **It encourages creativity:** Using little creativity, social networking sites, such as YouTube and Instagram, have allowed users to create a professional career out of social media. This includes people making and uploading videos to YouTube, as well as Instagram models.
- **It can be used to increase your visibility:** Social networking sites, such as LinkedIn, allow users to create a professional profile that can be viewed by potential employers. The site allows the employer to see your educational information, professional working experience, as well as your interests and achievements. Users can also upload their CVs and apply for job vacancies. Your online presence is also known as your digital footprint. Websites collect information as you use them, with or without the knowledge of the user. All the traces you leave online by way of your web activities make up your cyber profile.
- **It can be used to build a community:** Social networking sites allow users to find and create communities of people with similar interests, religious views, or nationalities.
- **It can be used to help a cause:** Raising funds to start a business, to help someone, or to simply try to get an education, has never been easier. Crowdfunding sites, such as GoFundMe, allow users to create a profile explaining their situation and why they are in need of funds. This then allows other users to donate money towards the project of their choice. Crowdfunding is a way of raising funds through customers, family, friends and investors online.

NEGATIVE EFFECTS OF SOCIAL MEDIA

As more people start depending on the internet and social media for their news and information, it is important to understand that what you put on the internet, might come back to haunt you.



Something to know

Since the Cambridge Analytica incident, Facebook has taken steps to improve its security to protect its users' information.



Something to know

Every time you click on a website, download a file, or update your Facebook status, you are busy creating a trail of virtual data that makes up your **digital footprint**.





Let's take a look at some of the following negative effects of social media:

- **It can create a false sense of relationships:** More and more people are building online relationships as it is much easier for them to meet and chat with someone over the phone or computer, than meeting in person. This can lead to a decrease in the number of real-life relationships, which can cause people to suffer from isolation and depression.
- **It can affect your privacy:** Social networking sites encourage users to share information about their personal lives, what they like and dislike, as well as where they live. This information can then be used by people with malicious intent, such as with the Facebook Cambridge Analytica case.
- **It can affect your productivity:** Social media sites, such as Facebook, contain information and entertainment that is tailored towards the likes and hobbies of users. This can cause users to spend a lot of time browsing the site for things that they find interesting.
- **It can lead to peer pressure:** One thing that has always been true is that humans like to be part of a group. Social media sites allow users to view the thoughts and opinions of the people in their friends' group. This could lead to users changing their opinions on certain topics due to not wanting to be left out.
- **It can affect your actions:** People tend to act differently on the internet than in real life. Some people are more confident while others might take advantage of the anonymity of the internet to bully and ridicule someone.
- **It can lead to the distribution of fake news.** Fake news is news, stories or hoaxes created to deliberately misinforming or deceiving readers. Usually, these stories are created to either influence people's views, push a political agenda or cause confusion and can often be a profitable business for online publishers. Fake news stories can deceive people by looking like trusted websites.



Activity 8.1

1. Write down the correct answer for each of the following questions.
 - a. Which of the following is NOT a social networking site?
 - A. Facebook
 - B. GoFundMe
 - C. Instagram
 - D. Google
 - b. Which of the following is a business networking site?
 - A. Twitter
 - B. WhatsApp
 - C. LinkedIn
 - D. YouTube
 - c. Which of the following social network platforms can be used to obtain an income?
 - A. Facebook
 - B. YouTube
 - C. Twitter
 - D. WhatsApp





Activity 8.1

2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 A social network that allows the user to share photos and videos with other users	A. Instagram
2.2 A social media and news platform that allows users to share their thoughts, ideas and opinions with other users, in 280 characters or less	B. YouTube
2.3 A video-sharing platform that allows users to upload and view videos over the internet	C. Facebook
2.4 A social networking site that allows users to donate money towards a cause or project	D. WhatsApp

3. Answer the following questions:
- Discuss the positive and negative aspects of social networking. Do you think that the good outweighs the bad?
 - Define the term "crowdfunding".
 - Define "anonymity" and explain why it can have a negative effect on online communication.

8.2 Impact of technology

In the introduction to this chapter, we took a look at how technology has influenced our daily lives. Smartphones help us to wake up in the morning, communicate with friends and family, and navigate our way around the world. Computers, on the other hand, can be used for everything; from the car's navigation system to entertainment. To better illustrate this, in this section we will be taking a closer look at some of the positive and negative effects that technology has had on society.



Figure 8.3: *Technology in our lives today*

POSITIVE EFFECTS OF TECHNOLOGY

We use technology for just about everything; including communication, travel, entertainment and education. Here are some additional ways in which technology has improved our lives:

- **Telecommuting (e-commuting):** Telecommuting or teleworking is a work arrangement made between the employee and the employer. The arrangement allows the employee to work from home, or from a location that is close to the employee's home by using computers, telephones and the internet. Telecommuting has been found to be advantageous to employers, saving them office space and improving worker morale saving travel costs and allowing a more flexible work schedule.
- **Improved healthcare:** There are many ways in which technology has improved our healthcare system. This includes the development of new medicine, new treatments and detection techniques, as well as the digitalisation of health records.
- **Enhances our ability to fight crime:** Technology not only helps us to fight; it also prevents crime. Examples of this include cameras, drones, facial recognition, online databases and metal detectors.
- **Increased productivity:** Regardless of the task you are doing, computers can make you more effective at it. For example, you can research any topic on the internet in less than an hour. Twenty years ago, you needed to travel to a library and then spend hours looking through books without knowing if any of the books would help you.
- **Making the impossible, possible:** Computers can complete tasks that are impossible or incredibly time consuming for humans to do. For example, in 1624, Henry Briggs published a book containing the logarithms for 35 000 numbers, which took him years to calculate. Today, a person with a computer could do the same work in less than five minutes.



- **Artificial Intelligence:** Artificial intelligence is important because it makes life much easier for everyone. The so-called weak artificial intelligences, such as the GPS in your car and the operating system assistants like Alexa, Siri, and Cortana, are all designed to make simple things like searching for something online or navigating your way through an unknown street much easier. On the other hand, strong artificial intelligences, such as intelligent robots, are used in space exploration programs, performing tasks that would be otherwise impossible to undertake by humans by themselves. They are also applied in the medical field to perform complex surgeries that would be impossible or extremely difficult for humans to successfully complete.
- **Machine learning:** The concept of machine learning allows computers to take in large amounts of data, process it, and teach themselves new skills using that input. Used for climate change modelling, autonomous vehicles, medical research and risk assessments. The concept of machine learning allows computers to take in large amounts of data, process it, and teach themselves new skills using that input.
- **Block chain:** Known as the backbone technology behind Bitcoin, is one of the current trends on the market. Due to the decentralized networks, blockchain does not have a central point of failure and is better able to withstand malicious attacks, users can trust that transactions will be executed exactly as they want and transactions can be processed 24/7.

NEGATIVE EFFECTS OF TECHNOLOGY

Technology has improved productivity, helped scientists to cure diseases, assisted engineers to design and construct fantastic new structures, and it has connected people from across the world. However, it is important to not be blinded by the advantages and ignore the disadvantages. Let's take a look at some examples of how technology might harm society:

- **Working environment:** Face-to-face meetings have often given way to video conferences, mailrooms to email inboxes, and typewriters and carbon paper to word processors. Technology has also allowed a substantial portion of work—and the workforce—to move beyond the confines of a traditional office.² It is common for digitally connected professionals to perform some of their work in cafés or shops, at home, even lying by the pool while on 'vacation'.
- **Environmental problems:** The eternal race between companies to improve and gain a competitive advantage over others, does unfortunately negatively affect the environment too. Examples of this include e-waste, the depletion of natural resources and increased pollution.
- **Social effects:** Computers and the internet have increased our ability to communicate with people around the world, but by doing so, they have also decreased our personal and social interactions. An example of this can be seen when looking at the Hikikomori people in Japan. These reclusive people have withdrawn from society and use computers and the internet to isolate themselves to their homes.
-



- **Crime and abuse:** Computers give anyone connected to the internet, access to an incredible amount of information. Usually, people use this information to keep up to date with current events, improve their existing skills, or learn new skills; however, there are some that use the information to commit terrible deeds. Examples of this include cyberbullying, online stalking and computer fraud.
- **Lack of upskilling:** as businesses are getting digitised at a faster rate, upskilling employees regularly has become a major factor. Most employees are concerned that their skills are either already outmoded or will soon become so.

INFORMATION OVERLOAD

Have you ever found yourself in a situation where you need to decide on a matter, but you just cannot make up your mind? It could be that you are suffering from *information overload*. Information overload is a term used to describe a situation where there is just too much information available for your brain to process. In short, the hard drive is full, but you are attempting to copy more data to the drive.



Figure 8.4: Information overload

Information overload is caused from our brains' limited capability when it comes to processing information, as well as the massive amount of information it is subjected to each day. In order to help you deal with this problem, let's take a look at some tips on how to deal with information overload.

VIRTUAL AND AUGMENTED REALITY

One of the most exciting and interesting advances of technology, is virtual or augmented reality. Virtual reality is a type of output device that uses a specially designed headset to fully immerse the user in a high-quality 3D virtual world, using both sight and sound.

What sets virtual reality apart from other 3D viewing devices is that it is *not* limited to just normal 3D images. The headset functions to adjust your vision, depending on where you look in real life, changing the picture you see as you look at different things.

This gives the device a much more immersive feeling. Due to this, there is a high level of concern about the potential negative effects that virtual reality might have on the user.



Let's take a look at some of the potential negative effects of virtual reality:

- **Addiction:** The immersive feeling of virtual reality might cause some users to turn their back on the real world.
- **Cybersickness:** This is a form of motion sickness that is caused due to using virtual reality technology.
- **Physical injuries:** Prolonged use of virtual reality can lead to eye strain and muscle fatigue.
- Virtual reality devices can be used for a variety of things; including training simulations and games.



Figure 8.5: *Using virtual reality goggles*

Unlike virtual reality, **augmented reality** does not fully immerse the user in a 3D virtual world instead, it adds digital elements to the real world. These elements can be viewed and interacted with by making use of a camera or smartphone. The most well-known example of augmented reality can be seen when looking at the super popular mobile game, "Pokemon GO". In this game, players use their smartphones in order to find and catch Pokemon in the real world. The game uses the phone's GPS to locate and move each player's virtual character on the game's version on Google Maps.



Activity 8.2

1. Answer the following questions:
 - a. What is a digital footprint and how can the average user obtain one?
 - b. When is a person described as a Hikikomori?
 - c. Are there any benefits to using e-commuting? If so, what are they?
 - d. Discuss three positive and three negative effects that technology has had on modern society.





REVISION ACTIVITY

QUESTION 1: SCENARIO-BASED QUESTIONS

Ntuli has been given a big project that needs to be completed within a few days. This project has left him stressed, because he has to do a lot of research. After doing his research, Ntuli found that he had approximately 326 pages worth of research to go through.

- 1.1 What could Ntuli suffer from as a result of this project? (1)
 - 1.2 What four things can Ntuli do to prevent or reduce this problem? (4)
 - 1.3 When Ntuli completes his school project, he is very proud of it. He decides he wants to share it online so that he can get people's opinions of his work. What could he use to do this? (2)
 - 1.4 Mention four things that Ntuli should be cautious of when sharing his project online. (4)
- [11]

QUESTION 2

Copy and complete the following table. Which of the statements in the table are *negative* side effects of social media, technology and virtual reality?

STATEMENT	NEGATIVE SIDE EFFECT OF ...
2.1 It can lead to peer pressure	
2.2 It can cause cybersickness	
2.3 It can create a false sense of relationships	
2.4 It can lead to crime and abuse	
2.5 It can lead to information overload	
2.6 It can affect your privacy	
2.7 It can lead to addiction	
2.8 It can affect your productivity	
2.9 It can affect your actions	
2.10 It can affect the environment	

[10]

TOTAL: [21]

AT THE END OF THE CHAPTER

NO.	CAN YOU ...	YES	NO
1.	Explain the impact that social media has had on the community?		
2.	Identify and list the benefits and harm of technology?		
3.	Explain what information overload is?		
4.	Describe the impact of virtual reality?		





TERM 1, 2, 3

INFORMATION MANAGEMENT: PRACTICAL ASSESSMENT TASK

CHAPTER 9

CHAPTER OVERVIEW

Unit 9.1 Problem solving cycle

Unit 9.2 Information management

By the end of this chapter, you will be able to:

- Discuss the different methods that can be used to collect data.
- Process and analyse data.
- Present data..

INTRODUCTION

Every day, we are presented with new problems and challenges. These problems can be simple, such as deciding what to have for breakfast, or more complicated, such as trying to find out why your car is not starting. To solve these problems, you need to find and process data so that you can do the following:

- Identify what the problem is or how the problem occurred.
- Find a possible solution.
- Find out how to implement the solution.

You can do this by following the steps in the information management process, shown in the diagram below.

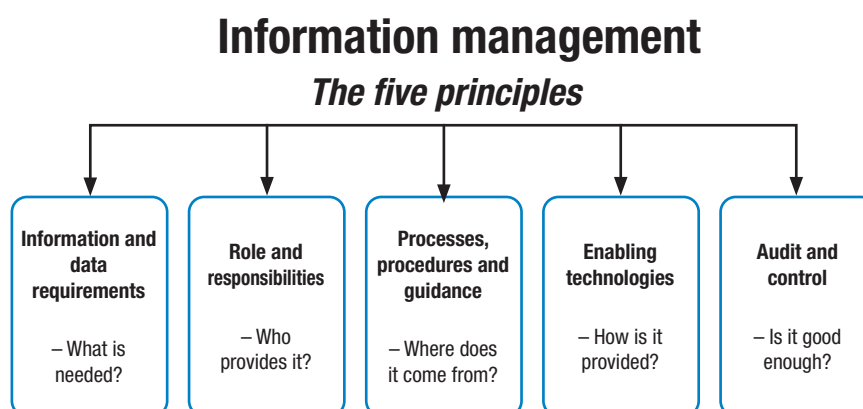


Figure 9.1: *The information management process*





The information management process is a good example of how to collect, organise, interpret and present data as information. To help you get a better understanding of this process, we will take a closer look at information management. We will start by looking at how data can be gathered. We will then discuss how data can be stored, how to check the quality of the data, as well as how to process the data. Finally, we will take look at how data can be interpreted to give us information and how we can present this information to others.



9.1 Problem solving cycle

In previous years, you learned that data is raw, unorganised numbers or facts and that these facts are difficult to use without being organised or changed. When data is organised so that it can become useful to people, it becomes information. Converting data into information can be seen when solving a problem. Problem solving is a process where you attempt to determine a solution to your problem by looking at the data and trying to organise it into useful information. Problem solving consists of the following five steps:

1. **Define the problem:** Here, you need to determine what the problem is you are trying to resolve. In order to do this, you need to understand what the problem is. The best way to do this is to write down everything you know about the problem. Make sure to include the following:

- What is known about the problem?
- What information is missing or is needed in order to determine the problem?

After you have written down everything that you know about the problem, you need to determine how you are going to find the missing information. One of the ways in which to do this is to use questions or questionnaires to collect data on the problem. This data can then be sorted into useful information that you can use to identify the problem as well as possible solutions.

2. **Identify possible solutions:** Once you have identified the problem, you need to research ways in which to solve the problem. You can use information sources and **data-gathering tools** for this. **Information sources** include electronic sources, such as wikis and internet articles, and printed media, such as books and standard operating procedures (SOPs). Data-gathering tools include conducting questionnaires and interviews with the relevant people or targeted market. Both of these sources will provide data that can be organised in order to come up with possible solutions.

3. **Choose a solution:** Your research would have identified several possible solutions to the problem, but how do you determine which solution is best? To do this, you need to apply your knowledge of the subject. Knowledge is something you gain through experience and education. Your knowledge of the subject should enable you to sift through all the information that you have obtained in order to identify the best possible solution to the problem.

4. **Implement the solution:** If you use your knowledge as well as the information that you have gathered, you should now have identified the best possible solution. Make sure that you implement the solution according to your knowledge and research. A solution that is not fully implemented might create new problems.

5. **Review the solution:** Problem solving does not end with the implementation of the solution. You still need to determine if the solution is effective for resolving the problem. If the chosen solution is not effective in resolving your problem, you will need to go back to Step 1 and identify why the solution did not work. Then repeat the process by choosing a new solution and testing to see if it resolves the issue.



Something to know

Sifting through information is the process of **ONLY** keeping information that will enable you to solve the problem.

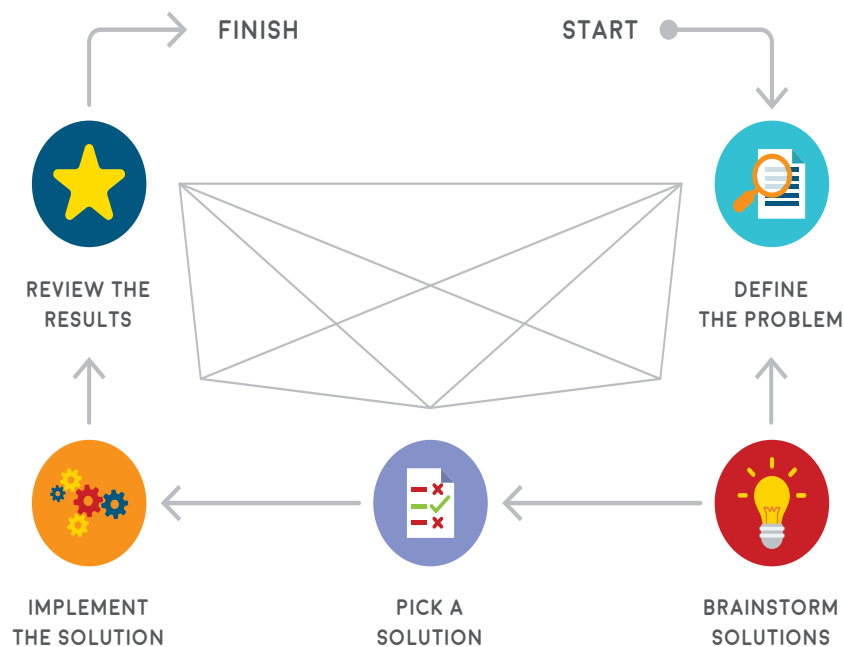


Figure 9.2: Problem solving

Example 9.1 Use data to solve a problem

Miss Peterson is a High School Natural Sciences teacher who is very concerned for her students. Not only does she want her students to pass; she wants them to strive to be the best that they can be. So, in order to determine which students are underperforming, Miss Peterson has decided to look at the school's database to see how her current class is performing when compared to previous year's classes.

When Miss Peterson opens the database, she finds hundreds of pages containing data for each student that has enrolled in the school over the last five years. Miss Peterson requests the database to convert all the data into percentages and to display only the data for students that take her Natural Sciences class.

Once the database has converted all the data, Miss Peterson is presented with five columns containing the averages for her students over the last five years. She can now use this data to compare the averages and identify the students who will need extra attention and support.



Activity 9.1

1. Write down the correct answer for each of the following questions.
 - a. Which of the following is part of reading an academic article on the problem?
 - A. Research
 - B. Knowledge
 - C. Experience
 - D. Work



Activity 9.1

... continued

- b. Which is the first step in problem solving?
- A. List possible solutions
 - B. Review the suggested answer
 - C. Find the answer
 - D. Define the problem
- c. What are raw, unorganised numbers or facts called?
- A. Information
 - B. Knowledge
 - C. Data
 - D. Wisdom
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 A data-gathering tool	A. Questionnaires
2.2 The process of only keeping information that will enable you to solve the problem	B. Acquire and retrieve information
2.3 Examples include academic journals	C. True
2.4 An organisation uses an information management process to do this	D. False
2.5 Problem solving can be a continuous process due to "Step 5: Review the solution"	E. Sifting through information
	F. Exams
	G. Information sources
	H. Information management

3. Answer the following questions:
- a. What must be included in a problem statement?
 - b. List the five steps of problem solving.
 - c. Give four examples of information sources.
 - d. Explain how an interview can lead to solving a problem.



9.2 Information management

Information management is the process of collecting, processing, and presenting data and information. The process consists of the following three main steps:

1. **Input:** This is the first step of the information management process and consists of identifying the main problems and collecting data.
2. **Process and analyse:** Once all the data has been gathered, it is converted to information by processing and analysing the data.
3. **Output:** This is the final step in the process and consists of presenting the information in such a way that it can be easily understood by other people.

The information management process is very similar to the information-processing cycle that we discussed in Chapter 1, with the main difference being that the information management cycle consists of three main steps and not five. To recap, the five main steps of the information-processing cycle include:

1. **Input:** The first step in the information-processing cycle is when the computer receives data from the user. The data can be either in the form of information or instructions.
2. **Storage:** The input data is then stored by the computer. This can be either permanently on a hard drive, or temporarily in the RAM (the computer's short-term memory).
3. **Processing:** Once the data is stored, the computer can start manipulating the data. This is done according to instructions programmed into the computer.
4. **Output:** The manipulated data can now be sent to an output device. This includes a screen that can display the data as well as other devices, such as a printer.
5. **Communication:** The final step of the information-processing cycle is communication of the information to other sources. This can be done between two computers, or over a massive network, such as the internet.

PAT AND THE INFORMATION MANAGEMENT CYCLE

During your Practical Assessment Task (PAT) examinations, you will be asked to research and present your findings on a given topic. To do this, you will use the information management process.

Start by identifying the potential problems highlighted in the given topic. These problems can then be used to generate questions that you will use in order to gather data. Once you have gathered the data, it has to be processed and analysed to obtain information. The final step of the examination is to take the information and present it in a format that others will be able to understand.

Examples of potential topics include:

- What is the impact of fake news on society?
- An investigation into the shortage of high-school pupils and university students studying in the scientific, technical/technology and engineering fields.

In order to assist with your PAT examination, we will be using the following practical example to demonstrate each step of the information management cycle.



Example 9.2 E-learning in South Africa

With the advances in computers and the internet, e-learning has started to replace traditional forms of education, such as printed textbooks. This form of education has become very popular in large parts of Europe and America, but has yet to do the same in South Africa.

As a current student, your task is to determine whether or not e-learning can be implemented in South African schools.

STEP 1: TASK REQUIREMENTS

In order to determine whether or not e-learning can be implemented in South African schools, you need to understand exactly what is expected of you. To do this, you will need to write down the task requirements by looking at the instructions given in your PAT assignment. You can do this in the following way:

- 1. Identify the main problem you have to solve:** Start by writing down everything you know about the problem and determine the main question that you would like to answer.
- 2. Identify a possible solution:** Determine what information you will need in order to answer your main question. This usually requires creating questions with answers that will let you solve your main problem.

You can expand on your task definition by using mind maps, bulleted lists, or any other method that best describes your plan of action.

Example 9.3 Find answers to a question

In the previous example, you were instructed to determine whether or not e-learning can be implemented in South African schools. In order to answer this question, we first need to look at what we know about the given topic, after which we will need to determine a possible solution to the question.

WHAT DO WE KNOW?

E-learning, or electronic learning, is a new method of education that uses electronic technology (computer and the internet) to present a course or subject. It is most commonly used over the internet. However, it can also be used in the classroom as a substitute for physical textbooks. This method of education has become very popular in Europe and America, but has yet to take off in South Africa. Why would this be?

ANSWER THE MAIN QUESTION

Now that we have determined our main question, we need to look at possible questions that will help us gather all the data and information needed to answer it.

Here are some of the questions that we will need to answer in order to solve our main problem:

- Does South Africa have the infrastructure for e-learning?
- Do people know what e-learning is?
- Would people make use of e-learning?



STEP 2: GATHER DATA

Once you have determined your task requirements, you need to start looking at ways in which to gather the data and information needed to answer your main question.

To assist you in gathering data, we will now take a look at some data-gathering techniques that you can use.

INTERVIEWS

An interview is a conversation between two people; the interviewer and the person being interviewed (interviewee). The interviewer asks questions that are then answered by the interviewee. Interviews are an example of a primary data source, because data is gathered and analysed by the person conducting the interview. They can be conducted in a formal manner, focusing on structured questions, or in an informal manner where the interviewer asks general questions. Examples of interviews used to collect data include the following:

- One-on-one interviews conducted in person
- Interviews conducted over the phone
- Group interviews

When conducting an interview, make sure that the questions are clear and focused on the topic that you would like to discuss. Interview answers usually include descriptions that are not numbers and cannot be computed; therefore, this data collected through the interview process.



Figure 9.3: *Conducting interviews*

QUESTIONNAIRES

A questionnaire is a research method that uses a series of questions in order to gather data. These questions can be **multiple choice**, or questions that require the respondent to give a short description. However, it is recommended to use multiple-choice questions, as people are more inclined to want to complete the questionnaire. Questionnaires should, therefore, not require long explanations or descriptions.

Examples of questionnaires used to collect data include the following:

- Surveys
- Opinion polls
- Multiple-choice tests



Figure 9.4: *Completing a questionnaire*

Let's take a look at the following guidelines for preparing a questionnaire:

- Decide what you are trying to learn from the questionnaire.
- Identify potential questions that will help you get the information that you need.
- Use closed questions (yes/no), as well as open questions (questions that require a description).
- Keep your questions short and simple to avoid confusion.
- Ask your questions in a way that will not guide people to a specific answer.
- Keep your questionnaire as short as possible.
- Test the questionnaire to ensure that it achieves what you are aiming for.

The data collected from questionnaires can be either quantitative or qualitative. Qualitative data is collected by looking at the answers provided. Quantitative data can be collected by assigning numerical values to the questions. An example of this can be seen in a survey that asks you on a scale of 1 to 5, how satisfied you are with the service you have received.

OBSERVATION

Observation is a method of collecting data where the researcher observes and documents participants performing a task in their natural setting. Unlike interviews and questionnaires, observation does not require the researcher to ask any questions; instead, it requires them to observe the task using their senses.

Examples of using observation to collect data include the following:

- Observing the interaction between a teacher and a student
- Observing a mechanic fixing a car
- Observing a train conductor operating a train

Observation can be used to collect both quantitative and qualitative data. Quantitative data is collected by looking at the amount of times a specific task is performed. Qualitative data is collected by looking at how the task is performed.



Something to know

When using observation to collect data, do not disturb or interrupt the person, or process, you are observing. Interruptions might lead to inaccurate observations as it might cause deviations in the process.



DOCUMENTS AND RECORDS

Written documents and records are one of the oldest methods used to capture data and present information. These documents can consist of databases, notes, reports, letters, records and written accounts. In the modern day, documents and records are stored electronically on your computer. The data that is obtained from documents and records is classified as secondary data. This is because the data was not generated by the person conducting the research, but rather by someone else. Examples of documents and records being used to collect data include the following:

- Looking at your financial records
- Examining the notes for a meeting that you did not attend
- Using information from a textbook



Figure 9.5: Documents and records

THE INTERNET

The internet has played a very big role in allowing people around the world to access a wealth of data and information. Using a search engine, such as Google, is much more time efficient and effective than visiting your local library.

Let's take a look at the following examples of where the internet is being used to collect data:

- **Wikipedia:** This website contains a wide variety of data and information. (Information obtained from Wikipedia should always be checked for quality and accuracy.)
- **PriceCheck:** This website allows you to compare the cost of items from different retailers.
- **LinkedIn:** This website provides information on job vacancies, as well as the professional profiles of people looking for part-time and full-time employment.



Figure 9.6: Using the internet to obtain data





Example 9.4 Further research

Unfortunately, the information obtained from the data analysis is not sufficient to make any final conclusions. In order to obtain additional information, we used our own data as a guideline and did further research using secondary information sources, such as the internet and written media. When you use secondary information sources, always check the quality of the information to ensure that it is valid and accurate. You can ask a variety of questions to check the quality of information.

Let's take a look at some examples:

- **Questions focusing on facts:** These type of questions aim to gather as many facts as possible to ensure that the information is accurate:
 - Who generated the information?
 - Is there evidence to support the information?
 - Is the information relevant and up to date?
 - How many times has the information been used?
 - Does the information cover your topic?
 - Is the information biased toward a specific outcome?
- **Questions focusing on exploring the information:** These questions aim to help you get a better understanding of the information:
 - What does the information indicate?
 - Why does the information indicate this?
 - How did the information come to its conclusion?
- **Questions focusing on predicting the outcome of the information:** These questions help you use the information to predict a possible outcome, or come up with a hypothesis:
 - Will I be able to use this information to find a solution to my problem?

The internet contains a wealth of information that has been uploaded by many different people; however, not all the information is accurate. This is because some people do not do proper research to ensure that the information that they put on the internet is accurate. There are also some that attempt to mislead you on purpose. Due to this, it is especially important to ensure that information obtained from the internet is accurate. Check the website that you are using to make sure it contains accurate and relevant information from a reliable and trusted source. You can do this in the following way:

- Make sure the website is easy to navigate.
- Determine for whom the website is intended.
- Check that the website and its information are up to date.
- Look at who supports the website.
- Check the credentials of the author of the information you are looking at.
- Check that the links on the website are working. This is especially important when checking references.
- Make sure that the information on the website is similar to that obtained from other websites.



Something to know

Sites, such as Wikipedia, can be updated by any person who can access the website. Information obtained from such a website should always be verified from another relevant source.





Something to know

Group your files and folders to make it easier to locate and use your data.

Example 9.5 Further research

In the previous section, we came up with five questions that we would need to answer to determine if e-learning can be implemented in South Africa. After looking at each of the questions, we decided that the best method to answer these questions would be a questionnaire to test the knowledge of students, parents and teachers around e-learning.

Let's take a look at the following example of our questionnaire:

E-learning

General overview of e-learning in South Africa

Questionnaire/Self-test

The purpose of this questionnaire/self-test is to obtain information about the knowledge of students, parents and teachers of e-learning, and its use in South Africa. Circle the appropriate letter on the sheet to indicate your answer.

1. Are you a student, parent or teacher?
 - A. Student
 - B. Parent
 - C. Teacher
2. Do you prefer physical (newspaper) or digital (internet) sources of information?
 - A. Physical sources
 - B. Digital sources
3. E-textbooks are more cost effective than printed textbooks. Choose one of the following:
 - A. Strongly disagree
 - B. Disagree
 - C. Agree
 - D. Strongly disagree
4. Why did you make the choice you did in Question 3?
5. Do you know of any South African educational institutions that use e-learning?
 - A. Yes
 - B. No
6. Do you own any of the following?
 - A. Smartphone
 - B. Laptop
7. Does your school have Wi-Fi available?
 - A. Yes
 - B. No

Figure 9.7: An example of a questionnaire

SPREADSHEETS

A spreadsheet is an electronic document that can be used to capture data in rows and columns.



Example 9.6 Capture data in a spreadsheet

The questionnaire regarding the use of e-learning in South African education was handed to and completed by 16 students, 27 parents and eight teachers. To help us get a better understanding of the data collected, we decided to capture the data by using a Microsoft Excel sheet. This was done in the following way:

1. Create a sheet called "Students".
2. Add headings for each of the questionnaire questions.
3. Capture the data for each questionnaire completed by a student.
4. Repeat Steps 1 to 3 for the questionnaires completed by parents and teachers.

Here is an example of our data captured in Excel.

	A	B	C	D	E	F	G
1	Response	Question 1	Question 2	Question 3	Question 5	Question 6	Question 7
2	1 Student	B	A	Yes	A	Yes	
3	2 Teacher	A	D	Yes	B	Yes	
4	3 Teacher	B	B	Yes	B	No	
5	4 Parent	B	C	No	A	Yes	

STEP 3: PROCESSING DATA

In order to convert your gathered data into usable information, you need to **analyse** the data. This can be done by sorting and grouping the data in a way that is easy to understand. Once the data has been sorted, you can use various formulae, functions and queries included in spreadsheets and databases to analyse the data.

Let's take a look at some of the tools that you can use to analyse your data:

- **Mathematical operators:** Excel and databases use standard mathematical signs, such as a plus (+), minus (-), multiplication (*) and division (/).
- **Formulae:** A data analysis can be performed using the following formulae:
 - **SUM:** This function calculates the total of a range of numbers.
 - **AVERAGE:** This function calculates the average of a range of numbers.
 - **MIN:** This function returns the minimum value from a list of values.
 - **MAX:** This function returns the maximum value from a list of values.
 - **COUNT:** This function counts the numbers in a list of values.
 - **VLOOKUP:** This function allows you to search for a value in the left-most column of a spreadsheet. If found, it returns a value.
 - **IF statements:** This function allows you to output text if a case is valid or false. For example, the formula `IF(A1<A2, "Yes", "No")` will give an answer of "Yes" when A1 is smaller than A2 and an answer of "No" when A1 is bigger than A2.

Database programs, such as Microsoft Access, use queries instead of formulae in order to analyse data. Queries do exactly the same things as the functions and formulae in Excel, but with less manual work.

Example 9.7 Processing data

In order to analyse the data we collected using the questionnaires, we transferred the data to an Excel spreadsheet. From there, we used the COUNTIF function to count the number of answers for each question.

Something to know

The method used to sort your data depends on the type of data. For example, numerical data (numbers) can be sorted from lowest to highest, while string data (words) can be sorted alphabetically.



Note

There are many more formulas not discussed here; refer to the PAT rubric for more information on the complexity levels for formulas (SPREADSHEET COMPLEXITY).

This was done in the following way:

1. Open your sheet containing the answers obtained from students.
2. Create a new table with headings for each answer.
3. Use the COUNTIF function to count the number of times a specific answer was given.
4. Repeat Steps 2 and 3 for the answers obtained from parents and teachers.

The screenshot shows an Excel spreadsheet titled 'Students - Excel'. The data is organized as follows:

	A	B	C	D	E
	For which grades?	Gr12	Gr11	Gr10	Gr9
1	Iqshaan	Y			
2	Natasha	Y	Y	Y	Y
3	Zain	Y	Y		
4	Amey	Y			
5	Nadia	Y		Y	
6	Allan	Y		Y	
7	Stef	Y	Y	Y	
8	Christina	Y	Y		
9	Naasir	Y	Y		
10	Count	9	6	4	1

Figure 9.8: Processed data

The screenshot shows the Microsoft Access 'Query Design' window for a query named 'Sheet1'. The design grid is as follows:

Field	ID	Students	Gr12	Gr11	Gr10	Gr9
Table:	Sheet1	Sheet1	Sheet1	Sheet1	Sheet1	Sheet1
Sort:						
Show:	☒	☒	☒	☒	☒	☒
Criteria:						

Figure 9.9: Query in a database

Using the information obtained from the database, we can now start answering some of the main questions:

- Does South Africa have the infrastructure for e-learning? We know that more than two-thirds of schools already have Wi-Fi available and that every person that was interviewed owned a device that can be used for e-learning. This would indicate that South Africa does possibly have the infrastructure available to implement e-learning.
- Do people know what e-learning is? Our data indicates that about half of the people know what e-learning is and that only one-third is aware that e-learning is currently being used in South Africa.
- Would people make use of e-learning? Slightly more than half of the people make use of a digital platform to obtain their information and when asked if they are interested in making use of e-learning, two-thirds indicated that they are.

STEP 4: ANALYSIS

Once data has been processed it is time to return your problem. Are you able to answer your main question? Based on information from other sources and the results of processing in spreadsheets and databases, you must formulate clear arguments in support of a solution to your problem.

An example would be “Most teachers strongly agreed (80%) that e-textbooks would be more cost effective while 40% of parents strongly disagreed. The introduction of e-textbooks would require an advocacy program to convince parents of the cost saving in using e-textbooks.”

STEP 5: PRESENT INFORMATION

Now that you have collected, analysed and processed the data, the last step is to present all the data and information that you have collected so that it is easy to understand the main focus and outcome of the research. You can use the following methods to do this:

- Use graphs and tables
- Create a presentation
- Create a website
- Write a report

GRAPHS AND TABLES

One of the best ways to present your data is to use a visual representation of your findings. Table users visualise trends and comparisons by allowing them to organise the data; graphs help to represent the information visually.

To insert a chart in Excel, you can do the following:

1. Select the range.
2. On the *Insert* tab, select the chart type.

Something to know

If you are unsure about which chart to use, you can use the recommended charts option.

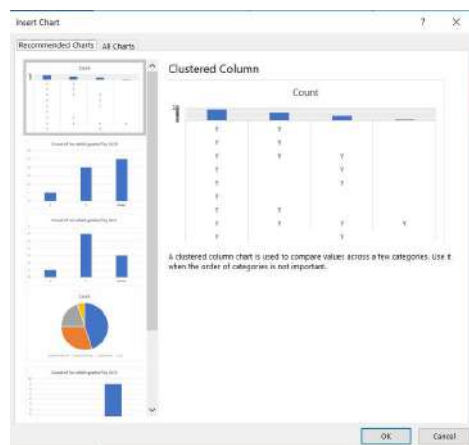


Figure 9.10: Inserting charts in Excel



PRESENTATIONS

PowerPoint presentations allow users to create a visual presentation of their findings through the use of slides. These slides can contain text, graphics and other multimedia sources, such as videos and animations. If you are looking to create a PowerPoint presentation in order to present your findings, here are a couple of tips:

- **Keep it simple:** Do not include too much information on a single slide.
- **Keep it short:** Try and present your information using as few slides as possible. This will keep your audience attentive and interested. However, do not leave out important information while trying to keep it short.
- **Do not use too many animations:** Excessive use of animations can cause the audience to get bored and lose attention.
- **Choose a font size and colour that are easy to read:** It is important to make sure that the audience will be able to read the text on your presentation, no matter where they are sitting. If the font is too small, or the text colour is unclear, many people might not be able to follow your presentation.
- **Use graphics and charts:** Instead of just having text, graphics and charts can be used to present your information in an easy-to-read and understand format.
- **Use the notes:** PowerPoint allows the user to make notes for each slide. This is especially useful to help you remember your main talking points for each slide.

CREATE A WEBSITE

During your PAT examinations, you might be asked to create a website to present your findings and information. This website will contain the same basic information that you would write in a report, with the difference being that the website will contain a more summarised version of the information. Follow these guidelines on how to create a website that is engaging and visually appealing:

- **Simple:** Simpler is better; websites should not contain too many design features, especially if those features serve no purpose.
- **Understandable:** The target audience of a website should be able to understand everything on the website without any difficulty.
- **Purposeful:** Websites should have a clear purpose, and every design and programming decision should work toward this purpose.
- **Useful:** A website should be useful and easy to use.
- **Consistent:** The design of a website should be consistent. This will make the website easier to understand and also make the website's identity clearer.
- **Visually appealing:** It is important to remember that even the most simplest, understandable, purposeful, useful and consistent website will not be used if people do not like looking at it.

High-quality websites should contain good information and have a good design.





REPORTS

Many people look at memory as a filing cabinet in your brain, storing pieces of important information until you need them. However, memories may also be inaccurate, or you may forget certain things. A written report ensures that you have an accurate record of all the relevant information.

A report is a form of communication that provides data and information of an event or incident. It is used to inform another person or party of the facts surrounding the event.

It helps us to clearly state the problem, the solution that was implemented to fix the problem, as well as how effective the problem-solving solution was.

A report can be written in an informal or formal style. An informal report is a short report; usually consisting of one to three pages. It provides information, but contains little to no research. It makes use of personal language and is normally very direct. A formal report is usually longer, contains facts obtained from the research and analysis of data, and is more direct.

WRITING A REPORT

The information in a report should be presented in a way that is easy to understand and to use. A report consists of the following sections:

- 1. Introduction:** The introduction is used to state the purpose of the report and provides a background on the issue. It is important to use your knowledge of the subject to provide a clear description of the problem that has occurred, as well as the solution that was implemented.
- 2. Body:** The body of the report will consist of a summary of information found from various sources such as the internet, books, surveys and so on. Information should be supported by graphs and other evidence such as the output of database queries. This will be followed by an analysis of the information you found. On the basis of the information found, a possible solution to the problem should be formulated. Any references to information sources should be acknowledged (cited). Care should be taken to avoid plagiarism by using your own words.
- 3. Conclusion:** This section is used to indicate whether or not the solution was a success and if any other solutions might be needed. If the solution was not successful, you need to include which steps you are planning to follow in order to make sure that the problem is resolved.

Always remember that it is important to write your report as soon as possible. This is to ensure that you include all the details that you have observed. The longer you wait, the better the chance that you might forget something important.



Something to know

Mid- to short-term memories are usually quickly forgotten. In order to help improve your short-term memory, you can use memory techniques, such as chunking. This is a technique where you take pieces of information and group them into larger units. An example of this can be seen when trying to remember a phone number.

For example, it is much easier to remember a phone number in chunks, such as 073 252 2103; instead of as a whole, such as 0732522103.



Something to know

Spreadsheets and databases can be used to organise and interpret any collected data by using tables and graphs. From these, the user will be able to determine trends and patterns that will provide information that can be used in your formal report.





Example 9.7 Present information

The final step in answering our question is to take all the information and data that we obtained from the questionnaire, data analysis and secondary information sources, and present it in a way that answers our main question. In order to do this, we decided to use a professional report.

A formal or professional report is an official report that contains a detailed data analysis, research, as well as the necessary information to make business decisions. Because of this, formal reports are usually used for problem solving. Examples of formal reports include the following:

- Inspection reports
- Safety reports
- Audits
- Annual reports

A brief, well-structured report that has clear objectives will get more attention and support, and is more likely to produce the intended results than a vague, poorly constructed report that takes a long time to reach the point. To help with this, here are the steps that you should take when writing a professional report:

1. **Determine the purpose of the report:** This is the first and most important step of writing a report. You need to decide what you wish to accomplish by writing the report. Make sure to get as much information as possible to help you determine exactly what you should include in the report.
2. **Determine for whom you are writing the report:** Once you know the purpose of the report, you need to determine who is going to read your report. This is very important as it will help you decide how much detail needs to be included, based on the experience of the readers.
3. **Write the report:** Now that you have identified the purpose of the report and who the report is aimed at, you can start writing the report. Make sure to have a clear, logical structure (introduction, body and conclusion) with clear headings to show where your ideas are leading. Take care not to make any assumptions about the readers' understanding and always explain why something is being said. Ensure that your report is long enough to fulfil its purpose, but not too long to lose the reader's interest.
4. **Review and revise:** The final step of writing any report is to review and revise what you have written. Make sure that everything makes sense and that the report is indicating what you had intended. It is recommended that you revise the report once some time has passed, for example a couple of hours later. This will allow you to review the report with a fresh mind set and will help identify mistakes that you might have missed previously.

THE STRUCTURE OF A FORMAL REPORT

In the first part of this section, we discussed the structure of a report, which includes an introduction, body and conclusion. However, a formal report may also include some additional sections. We will discuss these sections in three separate parts.

Part 1: Front section

This part of the formal report helps the reader understand what the report is about, as well as what is included in the report. This section contains the following:

- **Title page:** The function of the title page is to indicate what the report is about, who wrote the report, when the report was written and for whom the report was written.
- **Abstract:** An abstract is a summary of your report which you write after the report has been written. It should be written in the third person and in the past tense.





Example 9.7 Present information

- **Table of contents:** The table of contents gives the reader an overview of the document's contents. It also allows the reader to go directly to the section in which he or she is interested.

Part 2: Main section

This section contains the main report. It includes the following:

- **Introduction:** The introduction states the purpose of the report and provides background on the issue.
- **Body:** The body contains a discussion, analysis and provide possible solutions.
- **Conclusion:** In this section, the writer indicates whether or not the solution was a success and if any other solutions might be needed.

Part 3: Back section

The last section of the formal report is used to provide additional information that the reader might require. This includes the following:

- **References:** This is a list of the resources that were used during the writing of the report. The reader can use these references for additional reading, or to confirm the findings of the report.
- **Appendices:** The appendices include all documentation that was too large and bulky to be contained within the main report. This includes data sources, such as spreadsheets and databases. These documents are normally included to provide evidence for the findings of the report.
- **Glossary:** This is an alphabetised list of terms, definitions and abbreviations used in the report. The main function of the glossary is to provide the reader with a quick reference to terms that they might not recognise.



Something to know

When writing a report, take care not to plagiarise. Plagiarism is against the rules of any learning institution and could lead to you being prevented from finishing your studies. In order to show that you are responsible for obtaining the data and writing the report, you can use a declaration of authenticity.





Activity 9.2

1. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
1.1 Data that looks at the description of things	A. Tables
1.2 A recorded conversation between two people when one person mostly asks all the questions	B. Qualitative data
1.3 This helps the user to visualise trends and comparisons by allowing the user to organise the data	C. PowerPoint presentations
1.4 Part of the back section of a report	D. Interview
1.5 A way to present your findings to a crowd	E. Written report
	F. Keep it simple
	G. Glossary
	H. Make it useful

2. Answer the following questions:
 - a. List four guidelines to keep in mind when preparing a questionnaire.
 - b. Explain the difference between primary and secondary data.
 - c. What is the purpose of an executive summary?
 - d. Explain the three sections of a report.
 - e. Discuss the structure of a report. Make sure to elaborate on the three sections.
3. Which of the following criteria must be used to create a good presentation or website? State whether it is a presentation, website, or both. Copy the table into your exercise book and write the correct answer in the right-hand column.

CRITERIA	PRESENTATION/WEBSITE
Keep it simple	
Use graphics and charts	
Make sure it is understandable	
Keep it consistent	
Keep it short	
Ensure it looks beautiful	
Use the notes	
Do not use too many animations	
Make it useful	
Choose a font size and colour that is easy to read	





REVISION ACTIVITY

SCENARIO

Your local community has decided to run a campaign on healthy lifestyles. Amongst the activities they are organising is a triathlon race. They have asked you to help in sending invitations, creating documents that explain the event, recording results, keeping records of athletes and creating a website to market the event.

QUESTION 1: WORD-PROCESSING

Study the questionnaire below and answer the questions that follow.

Questionnaire: Living a healthy lifestyle.

A. Indicate your gender:

☐ M

☐ F

B. Why is it beneficial to live a healthy lifestyle?

C. Do you eat healthily? YES / NO

D. Do you love sport and living outdoors?

- 1.1 Identify any ONE closed question in the questionnaire. Explain why you say that the question is a “closed” question. (2)
- 1.2 Explain why it would be difficult to analyse the answers to Question B in a spreadsheet. (2)
- 1.3 Explain why Question D on the questionnaire is not a good survey question. (2)
- 1.4 The questionnaire does not have an introduction. Give ONE element that should be included in the introduction of a survey. (2)
- 1.5 The above questionnaire was done as a word-processing form. Explain how you would capture the responses to the questionnaire in a spreadsheet or database. (3)

[11]





REVISION ACTIVITY

... continued

QUESTION 2

In the PAT of 2018 learners were asked to investigate the problem of fake news. A learner asked the following question in her survey:

“How often do you see fake news?” Answer 1 to 5 where:

1 = Not at all

2 = Not frequently

3 = Sometimes

4 = Regularly

5 = A lot

How often see fake news , 1 to 5	
I less likely 5 most likely	Description
5	A lot
3	Sometimes
2	Not frequently
3	Sometimes
5	A lot
2	Not frequently
2	Not frequently
3	Sometimes
4	Regularly
4	Regularly
3	Sometimes
4	Regularly
5	A lot





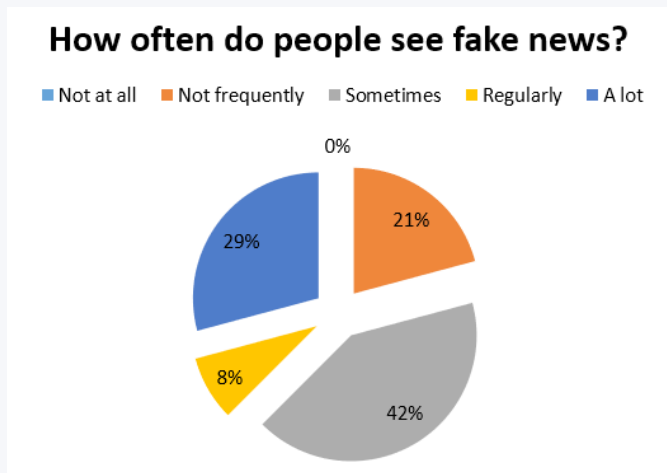
REVISION ACTIVITY

... continued

- 2.1** The learner captured the responses from each person who completed her questionnaire in a spreadsheet. Some of the results are shown in the image on the left. The respondents answered using the number 1 to 5. The learner used a spreadsheet function to convert the number to its matching description. What spreadsheet function did she use to do this? (1)

10. How often do people see fake news	
Not at all	0
Not frequently	5
Sometimes	10
Regularly	2
A lot	7

- 2.2** The learner then made a summary of all the responses to the question. What spreadsheet function did she use to do this? (1)
- 2.3** The learner created a graph of the results shown in 2.2.



- a.** Write an argument to include in a report based on the information in the graph. (3)
- b.** The number 0% appears on the graph but is not linked to a "pie slice". Why is this the case? (1)
- c.** What other types of graph could you use where it would be clearer which data item this is? (1)
- [7]



REVISION ACTIVITY

... continued

QUESTION 3

In the PAT of 2016 learners were required to investigate the problem of water shortages. A learner captured data about the capacity of dams which supply the Greater Cape Town area in a database table. The data spanned a period of three years from 2014 to 2015. The learner then created the following query:

Dam Name	Capacity (Ml)	Jan 11 2014	Jan 11 2015	Jan 18 2014	Jan 18 2015	Jan 25 2014	Jan 25 2015
WEMMERSHOEK	58644	90.6	76.3	90.6	76.3	85.2	71.1
STEENBRAS LOWER	31767	84.4	63.1	84.4	63.1	79.4	54
STEENBRAS UPPER	33517	98.8	99.3	98.8	99.3	99.7	96.6
VOELVLEI	164095	88.7	81.9	88.7	81.9	84.1	73.9
THEEWATERSKLOOF	480188	94	85.4	94	85.4	90.6	78.5
BERG RIVER	130010	98.6	88.1	98.6	88.1	99.5	80.8

The output of the query is shown in the figure above.

- 3.1 Suggest two query design techniques that will produce a meaningful outcome. (2)
 - 3.2 Write down the criteria to include all the dam names that starts with the letter 'S' and the dams that have exceeded capacity > than 80% in 2015. (3)
- [4]

QUESTION 4

As part of the PAT you are required to create a website. The image below shows a small part of the HTML.

```
<body bgcolor="darkblue">
  <h1><font color="darkslateblue">Load Shedding Facts</font></h1>
</body>
```

The image below shows the output on the browser screen.



- 4.1 What is the problem with the browser display? (2)
- 4.2 Make ONE change to the HTML above to correct the problem. (1)
- 4.3 A learner in your class says that if an image is on the internet, it is fine to use that image in your PAT website because it is for schoolwork. Is the learner correct? Justify your answer. (2)

[5]

TOTAL: 27

AT THE END OF THE CHAPTER

NO.	CAN YOU ...	YES	NO
1.	Collect data?		
2.	Process and analyse data?		
3.	Present data?		

Glossary

A

adware A type of virus that creates unwanted advertisements all over a user's computer. Adware is generally hidden in other installations and asks the user's permission before installing.

application software A type of software that allows the user to perform a specific personal, educational, or business-related function.

asynchronous communication A message can be sent without the message receiver being available at the time.

augmented reality Does not fully immerse the user in a 3D virtual world; it adds digital elements to the real world.

automation The process of using computers and software to automatically complete tasks with little human input.

AVERAGE This function calculates the average of a range of numbers.

B

blog An online diary that allows users to share their daily experiences with others.

boot process During this process, the computer checks that all the required hardware is in place and functioning, after which the operating system is loaded in the RAM. Once loaded, the operating system allows the user to interact with the computer hardware and software.

bot A software application that allows the user to automate tasks and actions. On the internet, bots are normally used to analyse and gather file information.

broadband A high data-rate connection that allows the user to send data over multiple channels. This makes the connection faster and more reliable than the old dial-up method. Examples of broadband include ADSL, 3G and fibre.

C

caching A small, temporary form of storage that is created in a fast form of memory in order to allow data to be accessed more quickly.

cloud-based file sharing An online file-sharing service that allows the user to share files that he or she has stored on the internet.

cloud computing A new technology that allows the user to store information, or use applications on the internet, instead of being on his or her own computer.

communication device A type of device that connects a computer to other computers in a network.

convergence A term used to describe a situation where multiple technologies are combined to deliver a new and more exciting product.

COUNT This function counts the numbers in a list of values.

Central Processing Unit (CPU) The primary component of a computer that processes data.

cyborg A being with both organic and mechanical parts.

D

data Raw, unorganised numbers, signals, or facts.

database An organised collection of data, usually stored in the form of structured fields, tables and columns.

database software Software designed to create and manage databases.

data-gathering tools Conducting questionnaires and interviews with the relevant people or targeted market.

data processing Involves using spreadsheets and databases to make data easier to understand and use.

denial-of-service (DoS) An attack that works by flooding a network or website with more information than it can handle, causing the network to be so slow that nobody can use it.

desktop PCs The most used type of computer.

disk cache Memory that stores data from the disk and allows fast activation of the disk.

distributed computing power A computing model that components of a software system and computer resources to be shared amongst multiple users.

document management software A type of program that allows the user to store, manage and track electronic documents and images.

E

e-filing Gives users a free, simple and secure way to complete and submit their tax returns online

email software Makes it possible for a user to compose and send email messages to other people using the internet, as well as to receive such messages.

encryption The process of converting data to an encrypted form that is protected from being viewed by unauthorised persons.

e-waste Any waste created by discarded electronic devices, components and substances used in their manufacture or use materials and heavy metals (lead, cadmium and



mercury) that are toxic.

F

fair-usage policy (FUP) A data limit placed on “uncapped” internet accounts. Your “uncapped” account will not be capped or limited unless you use too much data, in which case your internet access may be slowed (throttled) to the point where it is unusable.

files A collection of data stored in one unit, identified by a file name. It can be a document, picture, application, or any other collection of data.

file extension Consists of the letters that follow after the full stop in the filename.

file transfer protocol (FTP) A way to transfer data, particularly files, from one computer to another, usually over the internet but also over a local network.

folders The location where the files are stored on your computer.

fragmentation When you save a file onto your computer, the computer breaks the file up into smaller pieces in order to store the file on your hard drive.

G

gigahertz An indication of the processor’s speed. As a general guideline: the higher the frequency, the better the CPU.

green computing The study of designing, manufacturing, using and disposing of hardware, software and networks in a way that reduces their environmental impact.

grid computing A method of increasing the processing power at your disposal. It works by creating a computer network where each computer is able to use the other computer’s resources.

H

hotlinking Displaying an image on a website by linking to the website hosting the image.

I

IF statements This function allows you to output text if a case is valid or false.

information sources Electronic sources, such as wikis and internet articles, and printed media, such as books and standard operating procedures (SOPs)

instant messaging Describes online chat programs that allow the user to send real-time messages over the internet. This includes text messages, as well as

multimedia messages, such as pictures, voice notes and videos.

interdependent Hardware and software cannot exist without each other

J

keylogger A function that records keystrokes on a computer. It can be used by hackers or cybercriminals to steal your information.

keystroke logging A type of surveillance software that has the capability to record every keystroke made on a computer system.

L

laptops Portable PCs that combine the display, keyboard, processor, memory, hard drive and cursor positioning device (a touchpad or trackpad).

local area network (LAN) A computer network that covers a small area in which the computers in the network share resources, such as internet connections, printers and server connections.

M

malware Malicious software used by cybercriminals.

malware bots Infect the user’s computer and automatically perform malicious actions, such as gathering keystrokes, obtaining financial information, gathering passwords, or opening back doors on the infected computer.

MAX This function returns the maximum value from a list of values.

megapixels The resolution of the amount of detail that a camera can capture.

MIN This function returns the minimum value from a list of values.

N

netiquette Refers to internet etiquette. It is a list of rules and guidelines about acceptable behaviour on the internet.

P

personal computers (microcomputers) More commonly known as a PC; they are the smallest, least expensive and most used type of computer.

phishing and email spoofing attacks Methods used to obtain sensitive information (such as usernames, passwords and banking details) by sending emails that



look like official emails.

podcast A type of digital communication that uses an audio-only broadcast.

pop-up A form of advertising normally found on the internet

presentation software Specifically designed to help the user create and edit presentations in the form of a slide show. Presentations can include text, videos, or images.

protocol A set of rules for the transmission of data between devices.

R

read-only files Files that have been set; allowing the user to only read or move the file.

RSS A way to easily distribute headlines, updates and content many people at once.

S

scam A fraudulent scheme used by a malicious person to obtain money or something of value from someone else.

server A computer designed to process a large number of requests and deliver data to another computer over the internet or by using a local network

server computer A computer that can be accessed and used by multiple users through a computer network.

smartphones Handheld computers that use flash memory instead of a hard drive for storage.

sniffer attacks Data packets that are being sent over the network are inspected using a sniffer application, which allows the hacker to analyse the network and read all unencrypted data being transmitted. The information gained from the attack can then be used to crash the network, cause network data to become corrupt, or to observe network communications.

social networks The use of internet-based social media programs to connect with friends, family, customers and clients.

software A collection of data or computer instructions that tells the computer how to work.

software theft The unauthorised or illegal copying, sharing and usage of copyright-protected software programs.

spoofing attacks Attacks that are masked and makes it seem as if the attacker is a legitimate user. This is done by using falsified data to gain access to the network.

spooling A combination of buffering and queueing used by a printer.

spreadsheet software A type of program that sorts, arranges and analyses data in a table format. The user enters data into the columns and then does various calculations.

spyware Software installed on a computing device without the user's knowledge that is designed to gather data from that device.

storage capacity How much disk space a storage device provides.

storage speed How quickly new information can be written to the device or read from the device.

streaming A service that allows a user to play back a multimedia file without completely downloading it first.

stylus A small pen-shaped instrument used to directly press buttons and select options that appear on the computer screen.

SUM This function calculates the total of a range of numbers.

system requirements A list of hardware and software requirements that the software will need in order to function.

T

tablets A portable computer that uses touchscreen as its primary input device.

telecommunication Information that is obtained or exchanged using either a telephone, or a computer network.

typing tutor A type of software that teaches a user how to use the keyboard more effectively and accurately, as well as improve typing stamina and speed.

U

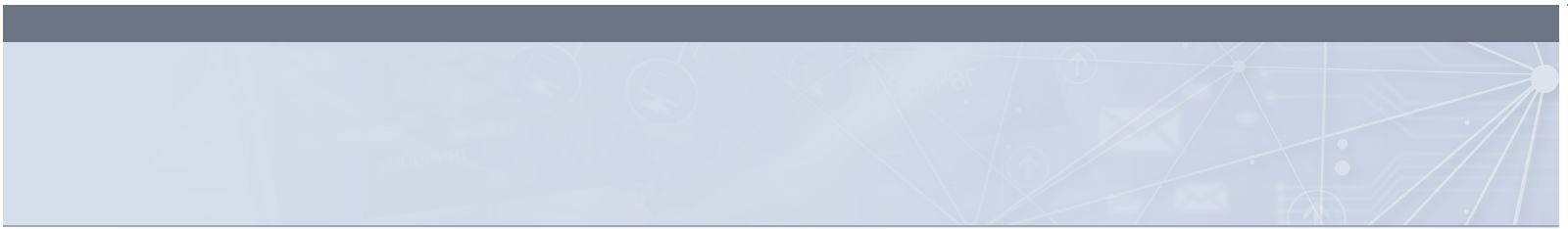
ultrabook Small and very powerful notebooks able to run any application. They have long battery lives and are easy to carry around.

user-centred design (UCD) Developers design software based on the focus and needs of the users.

V

vlogging Similar to blogging, with the major difference being that vlogging uses video instead of written stories and articles.

VLOOKUP This function allows you to search for a value in the left-most column of a spreadsheet. If found, it returns a value.



voice over internet protocol (VoIP) A telephone connection over the internet.

W

web-based applications Software that you can access over the internet.

web browser A software application that allows the user to access information on the internet.

web cache A small area on a hard drive where images and pages from the world-wide web are stored for a short period.

wide-area network (WAN) A network that covers a large geographical area.

wikis Specialised websites that allow users to share their information.

wireless access point (WAP) A networking hardware device that allows other Wi-Fi devices to connect to a wired network.

workstation A desktop computer typically networked and more powerful than a personal computer.

Z

zombie A hacked computer connected to a network. It can be used to perform malicious tasks that will compromise other computers on the same network.

